Thesis Report

Leveraging Unpaired Data to Mitigate Paired Data Scarcity and Foster Generalizable Multimodal Models

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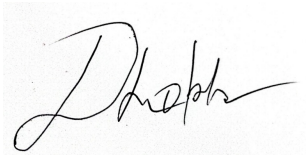
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Abstract

The increasing problem of antibiotic resistance calls for creative approaches to the discovery of novel antibiotics. In order to maximize biological activity, current AI techniques for producing antimicrobial peptides (AMPs) disregard the crucial practical limitation of synthesizability. The constraint may result in conceptually appealing peptides that are either prohibitively expensive or challenging to make, hence significantly impeding clinical translation. Our study bridges this basic gap by developing a revolutionary generative AI framework that may concurrently improve antibacterial efficiency and manufacturability using mismatched data sources.We propose a twostep method in which different models are trained with different knowledge bases: one for efficiency prediction and another for chemical principlebased synthesizability prediction. When generating new peptide sequences, these tell a limited variational autoencoder to automatically balance the two goals. We have shown that our methodology may provide over 1,000 candidate peptides with high levels of dualoptimization, compared to existing singleobjective approaches evaluated by computational tools. This study offers a revolutionary paradigm in computational drug discovery that could expedite the development of efficient treatment options against pathogen resistance by directly bridging the application/design interface between digital design and realworld implementation.

Keywords: Antibiotic resistance, antimicrobial peptides, AI drug discovery, generative AI, unpaired data, synthesizability, variational autoencoder, computational drug discovery, peptide design, dualoptimization, antimicrobial activity, clinical translation, dualcomponent model, digital design, practical application, drugresistant pathogens, peptide sequences, manufacturing constraints.

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Nomenclature

The next list describes several symbols & abbreviation that will be later used within the body of the document

$AAIndex$	Amino Acid Index Database
$AlphaFold2$	AI System for Protein Structure Prediction
AMP	Antimicrobial Peptide
AMR	Antimicrobial Resistance
$APD3$	Antimicrobial Peptide Database version 3
CNN	Convolutional Neural Network
$CVAE$	Conditional Variational Autoencoder
$DeepAMP$	Deep Learningbased Antimicrobial Peptide Predictor
DL	Deep Learning
$DRAMP$	Database of Antimicrobial Peptides
FFN	FeedForward Network
GAN	Generative Adversarial Network
GAN	Generative Models Using Adversarial Training
$LSTM$	Long ShortTerm Memory Network
MIC	Minimum Inhibitory Concentration
NLP	Natural Language Processing
$PULearning$	Positive–Unlabeled Learning
$RDKit$	Cheminformatics Software for Molecular Modeling
RL	Reinforcement Learning
RNN	Recurrent Neural Network
$ToxinPred$	Toxicity Prediction Tool for Peptides
$UniProt$	Universal Protein Resource Database
VAE	Variational Autoencoder

Chapter 1

Introduction

1.1 Background

AMR has emerged as one of the greatest twenty first century public health problems. The overuse and overprescription of antibiotics have led to the emergence of multidrug resistant bacteria, which have made current antimicrobial treatments less effective and have led to a rise in morbidity, mortality and healthcare costs globally. The World Health Organization (WHO) has declared antimicrobial resistance a major global threat to the successful treatment of many infectious diseases. With the emergence of antibiotic resistance, it is important to find alternative therapeutic agents that can be used to treat resistant pathogens.

Antimicrobial peptides (AMPs) are a promising class of next generation antimicrobial therapeutics. These are naturally occurring short peptides, that are synthesized by many organisms as part of their innate immune system. AMPs are usually not used to target a metabolic pathway as do traditional antibiotics, but instead disrupt bacterial cell membranes or interfere with several intracellular processes, making it much more difficult for bacteria to develop resistance.

The identification and optimization of antimicrobial peptides is a difficult process, even though they are very useful. Experimental screening requires a lot of laboratory resources, specialized knowledge and a significant amount of money.

In recent years, computational biology has been revolutionized by the development of artificial intelligence techniques such as deep learning and protein language models, which allow to predict biological properties directly from protein sequences. Protein language models like Evolutionary Scale Modeling (ESM2) are trained on vast amounts of unlabeled protein sequences to learn complex representations of protein sequences, which are hard to model with traditional handcrafted features and contain evolutionary and structural information.

While protein language models have shown tremendous success, sequence em-

beddings may not necessarily fully represent key physicochemical properties that affect antimicrobial activity. Peptide efficacy against bacterial pathogens is determined by critical biophysical properties including peptide length, molecular weight, hydrophobicity, net charge, isoelectric point, aromaticity and amino acid composition.

Encouraged by these developments, this study proposes a Biophysical Fusion Ensemble model to predict antimicrobial activity of peptides. The model we propose is based on a combination of pretrained ESM2 embeddings and handcrafted biophysical descriptors to create a detailed model of peptide sequences. Further, multiple independently trained neural networks are combined together by an ensemble learning strategy to enhance prediction robustness and estimate prediction uncertainty. The framework is validated by a carefully curated set of bacterial activity datasets, as well as by independent external benchmark datasets and internal evaluation, and is a reliable computational approach for the prediction of antimicrobial peptide activity.

1.2 Problem Statement

The emergence of antimicrobial resistance has made the need for effective methods of discovering new antimicrobial peptides against bacterial pathogens an urgent need. The identification and evaluation of antimicrobial peptides (AMPs) is still time consuming, laborintensive and costly in experimental settings, despite the potential of using AMPs as an alternative to conventional antibiotics. The vast number of peptides in the search space makes exhaustive testing in the laboratory impractical as well.

The use of machine learning and deep learning methods has greatly enhanced the computational prediction of antimicrobial peptide activity. Many current methods, however, only use handcrafted sequence descriptors or sequencebased neural representations, which are inadequate to represent complementary biological information necessary for accurate prediction. In addition, the predictive performance of the models can be poor when they are applied to peptides that have properties that are outside of what they have been trained on.

One further restriction is that many prediction models do not include uncertainty estimation. Confidence estimation is important for prioritizing peptide candidates for experimental validation, but is not well studied in many of the antimicrobial peptide prediction systems

Thus, it is necessary to develop a computational approach that incorporates both

protein language model embeddings and biologically relevant physicochemical properties, while also enhancing the prediction robustness by ensemble methods. This framework should be able to accurately predict antimicrobial activity, have reliable uncertainty estimates, and have a good generalization capacity in order to be applicable to different bacterial species.

1.3 Research Objective

General Objective

To build an AIbased system that predicts the antimicrobial activity of peptides against bacterial pathogens accurately, by combining protein language model embeddings with biophysical features.

Specific Objectives

1. To build a highquality bacterial antimicrobial peptide activity dataset by collecting, preprocessing, cleaning, normalizing and quality assuring data comprehensively.

2. To obtain informative peptide sequence representations with the pretrained protein language model Evolutionary Scale Modeling (ESM2).

3. To calculate biologically relevant biophysical properties of antimicrobial peptides.

4. To create a Biophysical Fusion neural network for the prediction of antimicrobial activity using ESM embeddings and biophysical descriptors.

5. To enhance the robustness of the prediction by adopting an ensemble learning approach with multiple independently trained neural networks.

6. Model performance assessment using multiple regression metrics, uncertainty estimation, novelty analysis and specieswise performance assessment.

7. To test the effectiveness and generalization ability of the proposed framework with independent external benchmark datasets.

1.4 Research Scopes

This study aims to create an artificial intelligence model to predict the antimicrobial activity of antimicrobial peptides against bacterial pathogens. The study

uses experimentally validated data on antimicrobial peptide activity obtained from public databases. These datasets were then comprehensively preprocessed, including sequence standardization, duplicate elimination, activity normalization, species harmonization, and data quality assessment, to guarantee the reliability of the datasets.

The proposed framework combines pretrained ESM2 sequence embeddings with handcrafted biophysical descriptors derived from the peptide sequence. The prediction is based on a Biophysical Fusion neural network using a fivemember ensemble strategy.

Common regression measures such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Fold Error, Pearson correlation coefficient, Spearman rank correlation coefficient, prediction uncertainty, novelty analysis, and specieswise performance evaluation are used to assess the performance of the model.

This research is limited to computational prediction and does not involve laboratory synthesis or experimental biological validation.

1.5 Thesis Organization

This thesis consists of five chapters.

Chapter 1 provides background on the research, discusses the motivation for prediction of antimicrobial peptide activity, and outlines the problem statement, research objectives, research questions, research contributions, scope of study, and overall organization of the thesis.

Chapter 2 summarizes the current state of the art in the field of antimicrobial resistance, antimicrobial peptides, AI in drug discovery, protein language models, deep learning methods for prediction of antimicrobial peptides, ensemble learning, and the research gaps tackled by this study.

The proposed methodology is detailed in **Chapter 3**, consisting of the following: data construction, data preprocessing pipeline, feature extraction, embedding generation of ESM2, computation of biophysical features, model architecture, ensemble learning strategy, training procedures, and evaluation metrics.

The experimental setup, results from internal evaluation, results from external benchmark, novelty based evaluation, specieswise performance analysis, and detailed discussion of the results obtained are presented in **Chapter 4**.

The research findings are summarized and the general conclusions are drawn in **Chapter 5**, while the limitations of the current research and recommendations for future research in the field of antimicrobial peptide discovery using artificial intelligence are presented.

Chapter 2

Literature Review

2.1 Preliminaries

Traditional AMP Discovery and Early Computational Methods:

The discovery of antimicrobial peptides (AMPs) has evolved significantly, shifting from conventional lab experiments to advanced computational methods. Early studies relied on isolating and analyzing peptides from natural sources, which led to the identification of key examples such as melittin and magainin. Early work established the importance of charge, hydrophobicity, and amphipathicity for antimicrobial function, paving the way for bioinformatics. With these in silico methods, researchers used motifbased searches and algorithms like SVMs and Random Forests to rapidly identify potential AMPs within large, curated databases. Early models had a major limitation as they could not generate entirely new peptides. They were limited to filtering existing sequences or making small adjustments, leaving most of the peptide space unexplored.

Rise of Deep Learning and Generative Models in AMP Design:

Deep learning and generative models like VAEs and GANs represented a quantum leap for the field. By learning complex sequence patterns, these systems could move beyond simple screening to the de novo creation of novel peptide sequences, venturing into previously unexplored chemical space. Studies in toptier journals confirmed the ability of these models to produce functionally relevant antibacterial peptides. The most recent innovation involves transformerbased models, which apply principles from natural language processing by treating sequences of amino acids as sentences in a biological language. This ability to model complex sequence relationships allows for the generation of highly accurate and diverse peptides. It marks a pivotal transition from passive screening to active design, substantially accelerating rational drug discovery by opening up a much wider range of candidate molecules.

The Gap: Synthesizability and MultiObjective Optimization:

While impressive, current models are hampered by a onedimensional focus on activity, neglecting the equally critical factor of synthesizability. A peptide's therapeutic potential remains theoretical unless it can be produced efficiently and costeffectively Current computational design frequently ignores practical synthesis challenges, including problematic length, difficult residues like cysteine and proline, and sequences that cause aggregation. A common workaround has been to use rudimentary filters after the fact to weed out nonsynthesizable options, but this is a blunt and wasteful solution that throws away many promising candidates The next critical step for the field is to pioneer true multiobjective optimization, developing models that are intrinsically designed to weigh therapeutic potential against practical manufacturability, a complex challenge that researchers are only beginning to tackle.

Future Direction: Bridging Efficacy and Practical Feasibility:

The next stage of AMP research should integrate synthesizability directly into generative modeling. Embedding chemical and structural constraints into the learning process would enable models to design peptides that are not only biologically active but also practical to produce. A particularly promising yet underexplored path is the use of unpaired datasets, where a model learns antimicrobial activity and synthesis constraints from separate sources. Mastering this approach would create a unified framework that seamlessly integrates biological and chemical knowledge, opening a significant avenue for innovationWe must now aim to unite AI's predictive power with pharmaceutical pragmatism, moving beyond theoretical designs to deliver peptide candidates that are both biologically compelling and industrially practica

2.2 Review of Existing Research

Current research into antimicrobial peptides (AMPs) is largely built on paired datasets that link peptide sequences directly to their labtested antimicrobial activity. These reliable datasets are key to training the sophisticated models that predict and create new peptides for medical applications. Szymczak et al. [1] created a model called HydrAMP which can make peptides with different levels of activity.They trained the model on labeled datasets containing peptides with known biological activity, enabling it to design sequences effective against bacteria yet safe for human cells. By understanding the link between sequence patterns

and antimicrobial activity, the system was able to design novel peptides and refine existing ones. In subsequent experiments, many of these designs proved to be powerful against bacteria yet harmless to mammalian cells. To narrow down the options, the researchers employed simulations to select the best prospects for further analysis However, HydrAMP was most effective with E. coli and shorter peptides of up to 25 amino acids. As a result, the authors recommended that future research include more structural and toxicity data, as well as tests on a wider range of bacteria.In another work, Jin et al. [2] made an AI system called AMPGen to design new AMPs. It used a diffusionbased approach together with ideas from evolution. In contrast with older systems that only looked at sequences, AMPGen also used multiple sequence alignments with evolutionary data to make better predictions. The model had three main parts: generator, discriminator, and scorer. It made peptides between 15 and 35 amino acids long. Before testing them, the researchers checked their chemical features and then analyzed them using tools like XGBoost and LSTM. When they made and tested 38 peptides in the lab, over 81% showed strong antibacterial effects, including some that worked on both E. coli and S. aureus. The system also made more diverse and realistic peptides than earlier ones. Still, there were problems with target selectivity and understanding how the peptides really function. The authors said future versions should include larger peptide families, better interpretation, and adaptive tuning for new diseases. Zhao et al. [3] also worked on AMP prediction and generation and built two systems these are called dsAMP and dsAMPGAN. The first system used deep learning to predict and classify AMPs, while the second employed a generative approach to create new AMPlike peptides. Both were trained on large datasets like APD3, dbAMP, and DRAMP, and transfer learning improved performance even with smaller datasets. The accuracy of dsAMP went over 95%, and dsAMPGAN made peptides that looked very close to natural ones. But since the data didn't have enough negative (nonAMP) examples, the performance dropped a bit. To address this problem, the authors proposed balancing the data or oversampling. Their findings revealed that labeled data can assist AI models learn how to create new peptides that match experimental results. Wang et al. [5] developed ProTDiff, combining diffusionbased methods with the pretrained protein language model ProT5 to generate realistic AMPs. The system included three connected modules for peptide generation, feature extraction, and activity prediction. After training on many AMP datasets, ProTDiff performed better than older models in originality and diversity. In lab tests, the peptides designed by this model proved to be highly effective at killing bacteria, confirming that the approach works in practice. Chung et al. [6] developed an ensemble deep learning model to predict the minimum inhibitory concentration (MIC) of AMPs against key WHOlisted

bacteria, including Pseudomonas aeruginosa, Escherichia coli, and Staphylococcus aureus. By combining eight different models (CNNs and BiLSTMs among them) with a weighted averaging method, their system achieved a Pearson correlation greater than 0.75, surpassing other models like ESKAPEEMICpred. This makes it particularly valuable for drug discovery, as it doesn't just label a peptide as active or inert but actually forecasts how strong its antimicrobial effect will be.

If summarized, these studies show very well how paired data can help AI make big progress in AMP research. By using both peptide sequences and their known activity, models can learn the link between biological function and sequence patterns. This helps them create new peptides that are safe and effective but still there are so many problems like data bias toward certain AMP families, short sequence limitations, and testing only a few types of pathogens. Also not having enough 3D or structured data makes predictions less reliable. Each group suggested future improvements, such as using more diverse peptide data, incorporating structural and toxicity information, and enhancing model interpretability. These steps lay a strong foundation for nextgeneration peptide drugs, and AIguided AMP design with paired data can make antimicrobial discovery faster, more efficient, and more accurate.

Nowadays, researchers have started using semisupervised and unpaired data methods to find antimicrobial peptides (AMPs). This helps when labeled data aren't enough or are missing. These methods let models learn useful biological patterns and make working peptides even without experimental labels like activity or toxicity. Ansari and White [10] worked on one big problem in peptide research. There are usually many peptides that show positive antimicrobial results but only a few that are confirmed as negative. The solution was a semisupervised system known as positiveunlabeled (PU) learning. This approach leverages a limited set of known positive peptides and a large, unlabeled set to identify critical features, including dissolution, coagulation, SHP2 binding, and nonspecific susceptibility Their approach includes a basic classifier that can determine which unidentified peptides are most likely not antibacterial. The model uses bidirectional LSTM recurrent networks to identify the patterns and amino acid structures that improve predictive accuracy. When evaluated on peptide datasets, this PU learning approach matched or even surpassed the performance of standard supervised methods, despite those methods requiring fully labeled positive and negative data. A key strength was its ability to reliably identify negative examples from the unlabeled data, which significantly improved the stability of the results. Additionally, the research showed that the model remains functional even in cases when the unlabeled data originated from peptide mutations. This is more in line with actual biolog-

ical processes. However, the authors noted that it is still difficult to obtain real negatives and that the system is heavily reliant on the caliber of the positive cases. They stated that more peptide properties should be studied and stronger model types ought to be utilized in future research. In general, our work demonstrated that even in the absence of labeled data, semisupervised learning can accurately predict peptide properties. You et al. [14] also followed this idea and made Dual Pseudo Training or DPT. IFor performing more effectively when there is a lack of data, it combines semisupervised learning with diffusion models. The same concept can aid in peptide discovery as well, but the main focus of their experiments were on picture collections such as ImageNet and CIFAR10. Two primary phases make up DPT. Masked Siamese Networks (MSN) or SemiViT are examples of semisupervised classifiers that first assign pseudolabels to unlabeled input. This gives a rough idea of what every sample could be.A conditional diffusion model makes new fake samples. This loop helps the model learn well from a small labeled set while still using big unlabeled data. In tests, DPT got great results like an FID score of 3.08 on ImageNet and about a 4.6% boost in accuracy even when less than 0.1% of data was labeled. This shows how models may function almost as well as fully supervised ones when pseudo labeling and data creation are combined. But there are still some problems. The models of diffusion are computationally demanding, and pseudolabels can introduce noise. It could get better in the future with the addition of filtering tools like CLIP or with improved pseudolabel quality. Although DPT was created for photos, the idea of creating synthetic data to support limited labeled sets can be extremely useful for AMP research, where it is difficult to obtain labeled biological data. Wan et al. [11] also made a big contribution with their model called APEX.The extinctome is a multitask deep learning system designed to analyze proteome information collected from extinct creatures. Discovering novel antimicrobial peptides was the aim. APEX employs distinct components for peptide classification and activities prediction, and combines recurrent and attentionbased networks to learn sequence characteristics. Large unlabeled proteome sequences in addition to labeled AMP data were used for training. The model studied more than 10 million peptide parts from 208 extinct species. It found 37,176 possible AMPs including over 11,000 that were totally new. When 69 peptides were made in the lab, 41 showed strong antimicrobial action and some worked as well as polymyxin B in animal tests. The researchers also discovered that peptides from the same extinct species generally performed better together than separately. Some even demonstrated high membranedepolarizing properties, which might aid in the development of novel antibiotics. The model has limitations. It didn't use 3D structure data and the extinctome data were small. But APEX showed how mixing labeled and unlabeled data can help

explore new biological sources and find more peptide candidates.

Together, the studies by Ansari and White [10], You et al. [14], and Wan et al. [11] demonstrate how semisupervised and unpaired learning enable effective AMP modeling despite limited biological dataThese methods help solve the lack of reliable negative samples and clear experimental labels. By mixing pseudolabeling, multitask learning, and data generation, these models can predict peptide activity, create new sequences, and find unique antimicrobial properties.However there are challenges remain, including data quality, high computational cost, and bias toward common peptide types. Future research should incorporate 3D structural data, larger and more diverse datasets, and closer integration between computational results and lab experiments. Overall, these studies highlight the move toward smarter, dataefficient AMP discovery capable of identifying novel peptides from rare or ancient biological sources.

In AMP research hybrid data methods are becoming more popular. These methods combine labeled data where peptide activity is known with unlabeled data from sources such as random sequences, genome files or computer generated samples. Using both types helps scientists predict peptide properties and create new sequences. Zhao et al. [3] built two deep learning models named dsAMP and dsAMPGAN. The first model predicts peptide functions using CNNs and BiLSTM layers. The second model called dsAMPGAN uses a GAN to generate new peptides similar to natural ones. Both were trained on large AMP datasets including APD3 dbAMP and DRAMP and used transfer learning to perform better with limited or uneven data. Tests showed that dsAMP got over 95% accuracy and dsAMPGAN made peptides with good charge, structure, and balance. Despite several issues, such as data imbalance and lacking biological information, their work showed that combining labeled and unlabeled data may significantly improve AMP prediction and production. Du et al. [7] also worked with big genome data and machine learning to find new AMPs in the Lactobacillaceae family. They analyzed over 10,000 genomes from 38 taxa and 515 species. Using models like APEX they identified 9,601 potential AMPs in 2,092 gene clusters. About 72 percent of these were completely novel and lab tests showed some could kill bacteria such as Staphylococcus aureus Escherichia coli and Pseudomonas aeruginosa. One peptide, delbruin1, was effective against various bacteria while being harmless for human cells. The study revealed how merging genetic data with lab tests might connect computer forecasts to realworld medical results. The scientists stated that more research is needed to prove the peptides' efficacy in various situations. Xu et al. [8] made another big step with a model called iAMPCN. It is a deep learning system that identifies AMPs and predicts 22 biological roles. It was trained on

a large dataset of over 49,000 peptide sequences from APD3 dbAMP DRAMP and LAMP. iAMPCN uses CNN layers and four types of encoding onehot BLOSUM62 AAIndex and PAAC to capture chemical and evolutionary information. The model works in two steps. A peptide's antibacterial, antifungal, anticancer, or antiviral properties are predicted after first determining whether it is an AMP. AMAP, AMPfun, and iAMPCA2L were among the older tools that the model outperformed. Accuracy and AUC values were over 0.99. Cysteine and lysine were identified as essential for peptide function based on amino acid research. Even though it worked great, Xu et al. [8] said it's still hard to tell apart similar functions like antibacterial and anticancer. They suggested using transformers and 3D structure data in the future to make it more accurate. Peng and Rajjou [4] worked on fixing AMP data quality. They found overlaps and errors in 28 AMP databases that reduced prediction reliability. They cleaned and merged them into a single standard dataset with natural and synthetic AMPs and negative samples from plant proteins They used tools like Dover Analyzer and CDHIT to remove repeated data and got over 50% nonredundancy.These cleansed datasets, which were posted to Hugging Face, improved the accuracy and stability of models. However, they added that it was more difficult to comprehend deeper biology when details like the synthesis type and peptide source were omitted. Their work showed that clean and hybrid datasets make AMP studies more reliable and repeatable. Dong et al. [9] took things further by creating a deep learning pipeline for finding bifunctional AMPs. Both paired and unpaired data were used in this approach. In order to assess antibacterial strength, their model combined a graphbased predictor with a GAN generator. They utilized AMP sequences to train one GAN and antiviral data to train the other. Out of 24 generated peptides, many had strong antibacterial and antiviral effects. Some peptides outperformed polymyxin B in animal tests. The top peptide P076 killed multidrugresistant Acinetobacter baumannii while remaining safe for mammalian cells. Simulations showed how these peptides bind to bacterial membranes and destroy them. Scientists noted that limited peptide length and incomplete knowledge of bacterial targeting remain challenges. They recommend using structural modeling in future research to better understand these behaviors.

These studies show that hybrid and mixed data approaches are transforming AMP research. Combining labeled and unlabeled data improves predictions, enables more creative designs, and addresses the shortage of labeled data. Hybrid methods let AI models generalize better, make more accurate predictions, and create new peptide types that actually make sense biologically. There nevertheless remain certain difficulties, such as adding 3D or environmental elements and maintaining

balanced and consistent data. Researchers will likely concentrate on more seamlessly integrating genetic, laboratory, and computer data in the future. Future models will learn from both structures and sequences, with lab validation needed to confirm computational results. Hybrid AMP research is speeding up the discovery of new antimicrobial and antiviral peptides by combining multiple data types with AI, helping combat antibiotic resistance worldwide.

The design of antimicrobial peptides (AMPs) is being revolutionized by recent developments in generative AI, namely diffusion models. These models facilitate the connection between biological investigations and computer forecasts. ProTDiff, created by Wang et al. [5], creates new AMP sequences from scratch by fusing diffusion modules with the pretrained protein language model ProT5. The goal was to fix old issues like small datasets, repetitive peptides, and low biological meaning. ProTDiff works in three main steps. ProTDiff first extracts features from amino acid sequences using ProT5. Trained on a range of antimicrobial peptide databases, the model's diffusion generator designs novel peptides and assesses their antibacterial strength. The resulting peptides proved effective in the lab and were notably more diverse and innovative than those from previous methods. Although the system required a lot of computational effort and occasionally replicated common peptide sequences, it was nonetheless a significant step forward for programmable AMP design. Lin et al. [12] advanced this approach with PepFlux, a physicsinspired diffusion model designed to generate peptides more like natural ones.To guarantee biological viability, the model integrated fundamental properties like charge and hydrophobicity. This grounding in realworld constraints enabled it to detect the specific, smallscale amino acid motifs that influence a peptide's antimicrobial action. While older models like GANs and transformers often struggled with stability, PepFlux consistently generated peptides that were both more stable and more novel. Its main limitation, however, was the lack of full 3D structural data in its design process. Soares et al. [13] also introduced OmegAMP, a diffusionbased tool that gives researchers more control over the design process. With this model, users can specify desired traits such as peptide length, charge, and hydrophobicity. It also uses a simplified system for encoding amino acids that still incorporates key physiological principlesOmegAMP used XGBoost classifiers and filtering methods to maintain diversity while improving AMP predictions. The model demonstrated superior accuracy and generated more diverse peptides than earlier tools (AMPGAN, DiffAMP, HydrAMP). However, its scope was limited to short peptides under 100 amino acids, and its performance was contingent on using highquality, curated datasets.Nevertheless, it raised the bar for biological clarity and user control in generative AMP research. The TGCDDPM model

from Cao et al. [15] is a textguided diffusion system that connects natural language to biological design. It works by generating peptides directly from textual descriptions of their target features. It links text and peptide data using contrastive learning and CLIP models.The workflow has three steps: matching text to peptide sequences, encoding peptide features, and generating new AMPs with diffusion methods. TGCDDPM was trained on over 116,000 textpeptide pairs and millions of synthetic samples. It made peptides that had very high activity scores (above 0.96), beating models like PepcVAE, ampGAN v2, and HydrAMP. Several of its peptides were shown in simulations to be able to attach to bacterial membranes and harm them. However, it did not include 3D structures and was restricted to short peptides, just as previous diffusion models. You et al. [14] worked on another method called Dual Pseudo Training (DPT). Originally developed for image generation, DPT adapts well to AMP discovery. It combines diffusion models with pseudolabeling, where a classifier trained on a small labeled set assigns labels to unlabeled peptides. These pseudolabeled peptides are then used by a diffusion model to create realistic new ones. The generator and classifier are both improved by repeating the process. By addressing the issue of insufficient labeled AMP data, this technique enables models to expand their training sets while maintaining biological sense.

Overall, these diffusionbased studies show a move toward smarter, more controlled AMP design. Models like ProTDiff [5], PepFlux [12], OmegAMP [13], and TGCDDPM [15] can shape peptide features precisely while maintaining real biological activity and better connecting AI predictions with lab testing. By mixing biophysics, protein embeddings, and control options, these models can make peptides that fit chosen goals like charge, stability, or target focus. However, there are many drawbacks, such as a high computational cost, short peptide restrictions, and a significant dependence on well selected data. Future studies could work with lab teams more closely, add more diverse data, and link diffusion models with molecular dynamics.

Recent advances in AI have significantly changed antimicrobial and therapeutic peptide discovery. Traditional methods depended on experimental screening or simple sequencebased predictions, which limited diversity and slowed progress. Wang et al. [16] proposed MMCD, a model that integrates peptide sequences and 3D structures using diffusion and contrastive learning. This approach improves peptide activity, diversity, stability, and docking affinity compared to traditional methods. MMCD predicts missing structures computationally and can guide broader peptide and protein design. Wang et al. [17] developed AMPDesigner, a large language model framework for rapid de novo AMP generation. It com-

bines pretraining, prompt tuning, model distillation, and reinforcement learning to produce biologically active peptides efficiently. Experimental testing showed that most generated peptides had strong antibacterial activity, with some performing exceptionally in lab and mouse studies. AMPDesigner also demonstrated adaptability using fewshot learning for limited data. Luo et al. [18] introduced AMPGFlowNet, a generative flow network guided by multiobjective reward functions. It generates diverse and potent peptides while balancing activity and toxicity. In silico evaluations confirmed antimicrobial activity and strong binding to bacterial membranes. Badrinarayanan et al. [19] proposed MultiPeptide, combining transformers with graph neural networks to leverage both sequence and structural data. The model achieved high accuracy in predicting hemolysis and nonfouling properties and improved latent space separation. Zhao et al. [20] developed EBAMP, a twostage framework using a Transformer for sequence generation and XGBoost for screening. EBAMP discovered broadspectrum antimicrobial peptides with low toxicity and strong activity against multidrugresistant bacteria and fungi. Experimental validation confirmed the safety and effectiveness of selected peptides. Collectively, these studies show that integrating sequence, structure, and property prediction enhances peptide design. AIdriven approaches allow exploration of larger peptide spaces faster and more efficiently than traditional methods. Multimodal and generative models produce diverse, biologically relevant peptides with reduced experimental costs. Future improvements include adding structural guidance, expanding datasets, and performing experimental validation to advance peptide therapeutics further.

2.3 Summary of Key Findings

Recent advances in artificial intelligence have greatly influenced antimicrobial peptide (AMP) research. Generative models and deep learning now help design novel peptides that fight resistant bacteria. A key advantage of conditional generative models is their controllability; researchers can steer the generation process to create peptides that possess targeted traits like potent activity or minimal toxicity.

i. These models learn deep patterns beyond simple sequence similarity.

ii. In laboratory experiments, a large number of AIgenerated peptides demonstrated genuine antibacterial properties.

iii. Classifiers and molecular simulations help rank and test the best peptides.

In AMP studies, a common issue has been the lack of data, particularly labeled

samples. Researchers employ hybrid and semisupervised learning to address this. Both labeled and unlabeled peptide data can be used in these models.

i. Positive–unlabeled learning helps when no negative data are available.

ii. Pseudolabeling allows the model to guess missing labels for training.

iii. Models can also incorporate genomic data or sequences from extinct species to discover new types of peptides.

Architectures like CNNs, RNNs, and Transformers are foundational to modern prediction tasks. Their integration into ensemble or hybrid systems often yields superior results, as each model compensates for the weaknesses of the others

i. They learn from amino acid patterns and physicochemical properties.

ii. Ensemble models increase accuracy and stability of predictions.

iii. These approaches don't only forecast "active" or "inactive" data; they also predict complicated data like minimal inhibitory concentration.

For AMP generation, diffusionbased models have gained popularity. They are able to modify specific traits and produce new sequences that adhere to biological laws.

i. Peptides can be generated with fixed length, charge, or hydrophobicity.

ii. Textguided diffusion enables designing peptide sequences using written descriptions.

iii. Multistep pipelines combine generation, filtering, and testing for better results.

Combining different data sources has improved model performance. Clean and hybrid datasets reduce redundancy and enhance learning. Mixing labeled, unlabeled, and synthetic data helps models generalize better. Removing duplicates and adding artificial negative samples improves balance. Mining genomes from rare or extinct species has uncovered many new peptides with strong activity and low toxicity.

Even with great progress, some challenges still remain. Models often struggle with larger peptides and need more computing power.

i. A major limitation in this field is the general lack of 3D structural data, which restricts a deeper biological understanding of how these peptides function.

ii. Data imbalance can bias models, causing them to overfit to wellrepresented AMP classes and overlook rare but potentially valuable types.

iii. A key priority for future work is to incorporate 3D structural information and strengthen the connection between computational designs and experimental testing.

In summary, deep learning has fundamentally transformed antimicrobial peptide discovery. By leveraging techniques like diffusion models, hybrid datasets, and semisupervised learning, researchers can now design and test novel peptides with unprecedented speed, diversity, and accuracy. AIdriven AMP discovery has emerged as a premier pathway for new antimicrobials, a promise grounded in the essential feedback loop between *in silico* design and *in vitro* validation.

Chapter 3

Impacts and Constraints

3.1 Final Specifications and Requirements

The suggested system is a deep learningbased computational system that will predict the antibacterial activity of antimicrobial peptides (AMPs) by estimating their Minimum Inhibitory Concentration (MIC) for a specific bacterial species. The proposed framework is fundamentally different from the conventional AMP prediction models because it is species conditioned MIC regression, which brings more biologically meaningful predictions for individual bacterial pathogens.

The system has two main inputs: antimicrobial peptide sequence and the bacterial species to be targeted. A pretrained protein language model called ESM2T68M is used to encode the peptide sequence as a 320dimensional contextual embedding vector, capturing the semantics and structures learned from millions of protein sequences. At the same time, 12 biophysical descriptors, such as molecular weight, net charge, hydrophobicity, hydrophobic moment, isoelectric point, aromaticity, instability index, aliphatic index, Boman index, charge density, GRAVY score, and peptide length are extracted, which characterize the physicochemical behavior of the peptide. Meanwhile, the bacterial species are coded in an embedding layer to record speciesspecific biological information.

They are all fused in the proposed Biophysical Fusion Architecture, and then fully connected regression layers estimate the logtransformation of the MIC value. An ensemble learning strategy is used to get more robust predictions by combining five models that have been trained independently, and prediction uncertainty is calculated as the standard deviation across the ensemble.

The final framework was implemented in Python with the PyTorch deep learning library and trained using NVIDIA T4 GPUs on Kaggle cloud. The model performance was assessed by internal partitions of train and validation and test sets, and external performance was conducted on the independent DRAMP dataset. The

evaluation metric adopted were MAE, MSE, RMSE, Fold Error, Pearson correlation coefficient, and Spearman correlation coefficient. The proposed framework is proposed as a computational decision support system for A.P. research. It will be used to identify peptide candidates for experimental validation and is not meant to supplant realworld testing.

3.1.1 Societal Impact

A growing resistance to antimicrobials, known as antimicrobial resistance (AMR), is one of the biggest health challenges in the twentyfirst century. Antibiotics are overused and misused, and many of the bacterial strains now exist that are resistant, meaning that many bacteria now become less responsive to the effects of antibiotics. Infections are becoming increasingly resistant, and costs to the health system, longer hospital stays, and mortality rates are adding up more and more for patients around the world. Thus, the development of new antimicrobial agents is a great need.

The proposed research is one of the solutions to this global challenge by creating an artificial intelligence framework that will predict quickly the antibacterial activity of antimicrobial peptides. The proposed system can be used to computationally test thousands of peptide candidates in a fraction of time to prioritize the most favorable candidates for experimental testing, instead of relying solely on time consuming laboratory experiments.

This computational screening approach can be beneficial to society in a few ways. It helps to save a lot of time at the initial phase of discovery of antimicrobial peptides. Second, it allows researchers to spend their lab time more efficiently by working on only those candidates which are highconfidence. Third, speeding up peptide optimization, the framework could be used for the development of future effective therapies against multidrug resistant bacterial pathogens. The proposed model is not directly designed to be used in clinical diagnosis or medical decision making; however, it is an important research tool that can assist biomedical scientists in their search for novel antimicrobial therapeutic agents.

3.1.2 Environmental Impact

Traditional antimicrobial drug discovery consists of intensive laboratory experiments with high consumption of biological materials, chemical reagents, disposable laboratory plastics and high energy consumption of laboratory equipment in continuous operation. Highthroughput experimental screening can also generate significant amounts of chemical and biological waste, and require a significant

amount of energy.

The computational framework proposed in this paper helps in improving environmental sustainability by minimizing the need for timeconsuming laboratory screening during the initial screening phase of the discovery of antimicrobial peptides. The researchers can virtually rule out a host of peptide candidates that are not worth further investigation because of their unfavorable properties, before they are subjected to experimental testing. As a result, less laboratory experimentation is needed, which not only helps to decrease the amount of reagents used but also helps to reduce the volume of hazardous waste, and the total use of resources.

While deep learning models require GPU computation and electrical energy to train them, the computation is still much more resourceefficient than screening thousands of peptide candidates. Moreover, GPU cloud computing services like Kaggle make efficient use of the hardware for several researchers to share resources, thus enhancing energy efficiency.

Thus, the proposed framework enables more sustainable biomedical research that includes the use of computational intelligence in the early stages of the antimicrobial discovery process, and minimizes the amount of laboratory experiments performed.

3.1.3 Ethical Issues

Ethical considerations are crucial for the responsible and transparent use of computational models in biomedical research, as AI is increasingly integrated into this field. While the proposed framework is concerned with the prediction of antimicrobial peptide activity and not direct patient care, some ethical issues are still relevant to design, development, evaluation and deployment of the system.

The ethical implications of the responsible use of artificial intelligence in scientific research are one of the key issues. The proposed model is designed for helping researchers to predict the antibacterial activity of Antimicrobial Peptides prior to laboratory testing. The predictions made should thus be taken as an estimate rather than a biological prediction. All peptides that are extremely active in the model need to be well validated in experiments to be taken into consideration for therapeutic development. If scientists take too much for granted from these calculations, they may draw incorrect conclusions and waste research efforts.

Data integrity and responsible use of a dataset is another ethical problem. The databases employed for this study are well known antimicrobial peptide databases

that are publicly available. During the preprocessing, a significant effort was devoted to eliminating duplicate records, to normalizing the names of the bacterial species, to standardizing the reported MIC and to removing biologically inconsistent records. Data quality is an ethical obligation since incorrect or inadequately managed data can lead to flawed models and predictions that could have negative consequences on future research.

The proposed framework does not involve any human subjects, patient data, or personally identifiable data. Therefore, there is no need for ethical considerations of informed consent, patient confidentiality and privacy protection in this study. Yet the management of publicly available scientific data is crucially important in order to comply with the licences and uses of the original biological databases.

Another ethical question that arises is algorithmic bias and fairness. Machine learning models are able to learn patterns in the data they are trained on. The model may be more likely to be accurate for overrepresented species or peptide classes in the training set and less likely to be accurate for underrepresented species or peptide classes. This can cause bias which can affect the results of future studies as the bacterial organisms that are well represented may be favoured. To address this concern, the proposed study built a biologically realistic bacteriaonly data set and tested the model with a number of different bacterial species as well as an independent external data set. While these measures help to increase fairness, future studies should further diversify datasets to allow for better prediction performance in a wider range of bacterial pathogens.

Ethical issues include transparency and interpretability. In the case of complex deep learning models, they are often "black box" models, meaning researchers may have trouble understanding the logic behind individual predictions. Although the proposed Biophysical Fusion Ensemble results in the accuracy of the predictions to be significantly improved, the decisionmaking process of the system is still not easy to interpret. To overcome this problem, the model architecture, feature engineering process, training method and evaluation method are clearly documented, making the research more transparent and reproducible. Further improvement of the model interpretability could be done by future studies using explainable artificial intelligence (XAI) techniques.

The other ethical issue is related to the responsible communication of model performances. Machine learning models should not be used to imply absolute accuracy, or that there is a 100% probability of them being correct. During the course of this research, a variety of different evaluation measures such as MAE, MSE, RMSE, Fold Error, Pearson correlation coefficient and Spearman correlation coefficient

have been used to report the performance. Additionally, independent DRAMP validation was carried out to give a more representative evaluation of model generalization. The inclusion of both positive and negative attributes of the proposed framework ensures that scientific integrity and responsible dissemination of research results is upheld.

Last, the possible societal consequences of faster discovery of antimicrobials should be taken into account. The framework could be used to create new antimicrobial treatments, but if the models are used incorrectly or if the peptides released are not sufficiently tested a negative effect may be achieved. Thus, the proposed system should not be regarded as replacing existing experimental, clinical research procedures but it should be considered as a computational decision support system.

Considering the whole, the proposed framework was developed following the principles of responsible AI, which relate to quality of data, transparency, fairness, reproducibility, and proper interpretation of computational forecasts. In conclusion, the ethical implications are integral to the safe and responsible use of AI in the development of antimicrobial peptides.

3.1.4 Standards

The implementation of the proposed antibacterial MIC prediction framework was developed according to the internationally accepted standards and good practices in the field of artificial intelligence, software engineering, machine learning and computational biology. By following accepted norms, the proposed system can be reproduced, reliable, maintained and scientifically believable.

The implementation was done in Python, a programming language that is widely used for scientific computing and machine learning research, and has a large ecosystem of opensource libraries. PyTorch offers flexible neural network building, automatic differentiation, GPU acceleration, and reproducible model training, making it an ideal framework for deep learning model development. The numerical computing, data preprocessing, and model evaluation were carried out using supporting libraries like NumPy, Pandas, and Scikitlearn, as per the scientific programming best practices.

We used the pretrained ESM2T68M protein language model as a stateoftheart method to generate protein sequence representation. Instead of relying only on handcrafted sequence features, ESM2 learns protein language representations from massive protein databases, it is one of the most popular protein language models

for computational biology research. Using ESM2 links the proposed framework to current international protein representation learning research efforts.

We have also adhered to the conventional practices of supervised machine learning model development. The dataset was split into three separate subsets: the training set, the validation set and the testing set, to avoid leakage of information and to provide an unbiased evaluation of the model. The selection on the model was based on the validation performance and the final evaluation was only done using previously unseen test data.

This approach is similar to standard machine learning assessment techniques in both research and industry.

The evaluation metrics were performed by the widely used regression evaluation metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Fold Error, Pearson correlation coefficient and Spearman correlation coefficient. Using several complementary evaluation metrics allows one to obtain a detailed picture of the accuracy of the predictions, their ranking consistency and their biological relevance, which increases the validity of the presented results.

From a software engineering point of view, the project focused on reproducibility and modularity. Each step of the dataset processing was developed in a separate module, enabling the independent reprocessing and verification of each step of the workflow. The hyperparameters, data preprocessing tasks, data partitions, and model design were meticulously recorded throughout the research to enable future replication and extension.

Good data management principles were also incorporated into the research. All peptide sequences were converted to a standard sequence before training the model, all bacterial species names were normalized to avoid semantic issues, duplicate records were eliminated, and all MIC values were transformed to a uniform representation. The preprocessing procedures increase the level of data integrity and minimize variation in the data caused by inconsistent experimental reporting.

While the framework is designed for research use and not for clinical application, it stems from guiding principles for the quality of AI, such as robustness, transparency, reliability, and reproducibility. To boost the confidence in prediction results, uncertainty estimates were added using ensemble learning, which provided increased stability in prediction results. External validation performed with an independent dataset of DRAMP showed further the credibility of the proposed framework since it was capable of generalizing the results obtained from the

training dataset.

Last but not least, the overall research process was conducted according to the standard of scientific research and academic integrity. There was good adherence to the use of prior research, reporting of the experimental method, objective evaluation of performance, and expression of the limitations of the model. The practices ensure the proposed research meets the expectations of reproducible, ethically conducted scientific investigation.

The proposed framework adheres to principles of artificial intelligence, computational biology, software engineering, and scientific research, ensuring a strong foundation for future development and potential integration into larger antimicrobial peptide discovery pipelines.

3.1.5 Project Management Plan

Phase	Duration	Activities	Deliverables
Phase 1	Weeks 13	Literature review on antimicrobial resistance, antimicrobial peptides, protein language models, and deep learning techniques. Identification of research gap and definition of project objectives.	Research proposal, literature review, problem statement, research objectives.
Phase 2	Weeks 46	Collection of AMP datasets from multiple public repositories. Dataset integration, preprocessing, MIC normalization, duplicate removal, sequence cleaning, and bacteriaonly dataset construction.	Final bacteriaonly dataset (32,607 records), trainvalidationtest datasets.
Phase 3	Weeks 78	Feature engineering, extraction of twelve biophysical descriptors, generation of ESM2 embeddings, and preparation of model inputs.	Processed feature matrices, cached ESM embeddings, feature extraction pipeline.
Phase 4	Weeks 911	Development of baseline, ESMbased, Transformer, and Biophysical Fusion regression models. Hyperparameter tuning and architecture optimization.	Individual prediction models and optimized Biophysical Fusion architecture.
Phase 5	Weeks 1213	Implementation of ensemble learning strategy, uncertainty estimation, and model integration.	Final Biophysical Fusion Ensemble model.
Phase 6	Weeks 1415	Internal performance evaluation, external DRAMP benchmarking, novelty analysis, and specieswise performance analysis.	Experimental results, evaluation reports, comparison tables, performance visualizations.
Phase 7	Weeks 1618	Thesis writing, documentation, revision, presentation preparation, and final project submission.	Complete thesismanuscript, presentation slides, and final defense materials.

Table 3.1: Project Management

3.1.6 Risk Management

In the field of engineering, especially in artificial intelligence, risk management is a crucial aspect of research, as it involves managing potential risks, uncertainties, and challenges within the development process. During the planning and implementation phase several potential risks were identified during this research, both technical and project related. The effect of these was minimised through appropriate mitigation strategies.

Data quality and data inconsistencies were one of the major challenges faced in the course of this study. The duplicate sequences, inconsistent bacterial species names, lack of activity data, nonstandard MIC units, and the presence of different biological targets (antibacterial, antifungal, antiviral, antiparasitic) in public antimicrobial peptide databases are not uncommon. These discrepancies can greatly affect the model and cause a loss of biological understanding. To address this risk, an extensive preprocessing pipeline was created to eliminate duplicate records, normalize bacterial species names, normalize MIC measurements, include only antibacterial records, and chemically validate peptide sequences before model training.

One of the major challenges was the issue of "overfitting" with the model. A deep learning model that has a large number of parameters that can be trained could memorize the training data instead of learning patterns in biology. This can lead to suboptimal performance on peptides that are not encountered before. Various mitigation methods were used such as separate training, validation, and testing datasets, dropout regularization, and early stopping based on the results obtained from the validation set, and ensemble learning methods. These methods enhanced the robustness of the models and helped prevent overfitting.

Another issue was the computational resources available. Protein language model based neural networks need a lot of computing power and GPU memory to train. To overcome this challenge, development and training of the model was done on NVIDIA T4 GPUs offered via the Kaggle cloud platform. In addition, ESM embeddings were cached to prevent redundant calculations in the model training process, which greatly reduced the amount of computation.

Another key project risk is the reliability of model predictions. Deep learning regression models tend to make nonrobust predictions that do not include confidence intervals. When analyzing new sequences of peptides, such predictions may be misleading. In order to lower this risk, an ensemble learning method was employed with five classifiers. The ensemble provides an estimate of the prediction

uncertainty (standard deviation of ensemble members) and therefore an indication of prediction confidence, in addition to improvement in predictive accuracy.

An external validation issue was also a potential threat. The best performing model on the internal test set may not perform well on external test sets. The risk was reduced by evaluating the proposed framework with the use of an external benchmark from the DRAMP database. This independent validation gave better proof of the generalisation of this model over the training set.

3.1.7 Economic Analysis

Because of the high cost of traditional drug discovery, there has been an increasing interest in the economic aspects of computational drug discovery. The traditional discovery of antimicrobial peptides has been a laborious process, involving a large amount of laboratory experiments, costly equipment, biological reagents, technical staff and repeated experimental validation. These activities make a significant part of the costs and timelines of the drug discovery process.

This is a proposed computational framework that offers an economical alternative to the evaluation of antimicrobial peptides at the early stages. Researchers do not have to try each and every candidate peptide to test it experimentally to see if it binds to the target; they can simply perform virtual screening on the developed artificial intelligence model. This computational prescreening allows for the identification of potential peptide candidates in a short time frame and the removal of candidate peptides with low predicted antibacterial activity. Accordingly, laboratory experiments can be targeted to a much smaller number of peptides of highconfidence, and hence laboratory investments are significantly limited.

The proposed framework is also effective in reducing the cost of software development. All of the system was built using opensource tools such as Python, PyTorch, NumPy, Pandas, Scikitlearn and the pretrained ESM2 protein language model. These are all available for free, avoiding software license costs and offering cuttingedge deep learning abilities.

Moreover, the cost of computational infrastructure was significantly reduced by leveraging Kaggle's cloudbased GPU platform that offers free access to NVIDIA T4 GPUs for academic research. This meant there was no need to acquire dedicated highperformance computing equipment and still be able to train models efficiently.

From an operational point of view, the proposed framework also enhances the productivity of researchers.

After training, the model can predict the MIC value of thousands peptide candidates in a matter of a few days, significantly lowering the time needed to prioritize peptide candidates. The fast computation screening could help researchers test their hypotheses quickly and speed up the entire research process.

Computational prediction will not eliminate the need to conduct experiments, but it will reduce the need for experiments considerably. The most costly part of the antimicrobial peptide development process is typically laboratory testing, reducing the number of unnecessarily conducted experiments directly translates to cost savings.

Aspect	**Traditional Screening**	**Proposed AI Framework**
Initial candidate evaluation	Laboratory experiments	Computational prediction
Screening speed	Slow	Highthroughput
Laboratory resource consumption	High	Low
Software licensing cost	Moderate	None (Opensource)
Hardware requirement	Laboratory equipment	Cloud GPU
Candidate prioritization	Manual	Automated
Overall research cost	High	Significantly reduced

Table 3.2: Comparison between traditional antimicrobial peptide screening and the proposed AIbased framework.

Chapter 4

Methodology

4.1 Framework

In this study, an artificial intelligence based model will be developed to predict the antimicrobial activity of antimicrobial peptides (AMPs) against bacteria species. It is based on an ensemble deep learning architecture that combines protein language model embeddings with handcrafted biophysical descriptors. The main goal is to achieve accurate prediction of the log10 MIC for peptide sequences, with the added benefit of making the model more robust and generalizable to various bacterial species.

The proposed framework has several sequential stages starting with the collection of the datasets and ending with the prediction of the activities and the evaluation of the model.

The overall methodology starts with collecting experimental data for the activity of AMPs from publicly available databases of biological activity. Data from various sources is integrated, and much preprocessing work is done to ensure data consistency, remove duplicate data, standardize activity measurements, normalize bacterial species names and eliminate invalid peptide sequences. This gives rise to a dataset that is used for training models.

After preprocessing, each peptide sequence is represented in two complementary feature representations. The first representation are protein embeddings extracted from the context, using the Evolutionary Scale Modeling (ESM2) protein language model. ESM embeddings are different from traditional sequence encoding approaches by incorporating contextual relationships between amino acid residues and learning structural and evolutionary information from millions of protein sequences. These embeddings are semantically rich representations of peptide sequence without needing explicit structural information.

The second representation is a set of manually designed biophysical descriptors

calculated directly from the amino acid sequence. These descriptors are important physicochemical properties that affect antimicrobial activity such as peptide length, molecular weight, hydrophobicity, net charge, isoelectric point, aromaticity, instability index, aliphatic index, Boman index, hydrophobic moment, and other sequencederived biochemical properties. The handcrafted features complement the sequence embeddings as they capture the physicochemical interactions between the antimicrobial peptides and the microbial membranes involved in their biological functions.

To incorporate speciesspecific antimicrobial activity, each target bacterial species is encoded using a learnable embedding layer. This allows the model to learn peptide susceptibility differences between bacterial species and the species information to be learned jointly with peptide representations during training.

The extracted ESM, normalized biophysical descriptors and bacterial species embeddings are then fused in a Biophysical Fusion Activity Expert. Different branches of the neural network are dedicated to each type of feature and are then fused into a single latent representation. Then, fully connected layers learn the nonlinear relationships between the characteristics of peptides and the antimicrobial activity, yielding a continuous prediction, which corresponds to the log MIC value.

This research uses an ensemble learning approach to better stabilize the prediction and decrease the variance of the model, instead of relying on a single predictive model. Multiple independently trained Biophysical Fusion Activity Experts are built with different random initialization seeds, with the same network architecture and training parameters. The activity estimate for the inference is computed by averaging the predictions made by each individual ensemble member. Further, prediction uncertainty is calculated as the standard deviation of the ensemble output, giving an indication of the confidence of the models in their prediction of the same.

The training is done on the peptide–species pairs, which are experimentally validated, and the dataset is split into training, validation, and testing sets. The model parameters are optimized by minimizing the loss function known as Mean Squared Error (MSE) between the predicted and experimentally measured log10 MIC values during optimization. Model checkpoints are continually evaluated for validation performance to ensure that overfitting is avoided and to determine the bestperforming checkpoint.

The proposed framework is then assessed through internal and external validation procedures after the training. Internal evaluation consists of assessing the predic-

tive performance of a regression model on the test set that is not used for training by applying regression metrics like Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Pearson correlation coefficient and Spearman rank correlation coefficient. External evaluation is performed by an independent benchmark set of peptides from DRAMP to test the model's ability to generalize to unseen peptides sequences while maintaining species specificity.

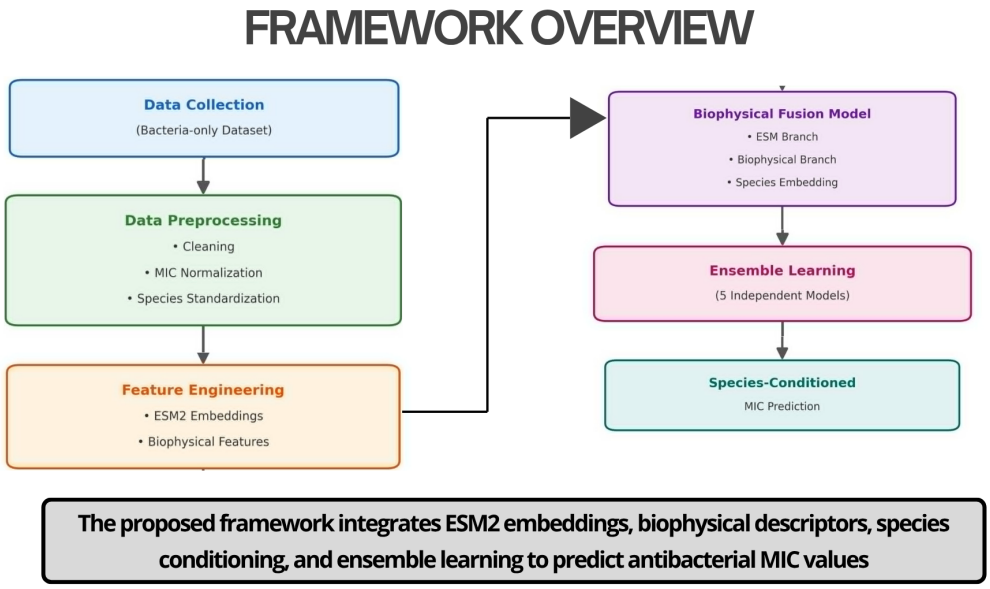


Figure 4.1: Framework Overview

The entire approach combines ensemble prediction, speciesaware learning, biologically relevant physicochemical feature engineering, and current protein language modeling into a single pipeline for antimicrobial peptide activity prediction. The suggested approach aims to improve computational APD prediction accuracy, robustness, and usability.By combining complementing biological data sources, the suggested methodology aims to increase computational APD prediction accuracy, robustness, and usefulness.

4.2 Data Curation

The quality, diversity and reliability of the data used for training is an essential determinant of the performance of any deep learning model. One of the most important aspects of this study was careful dataset construction, as the prediction of antimicrobial peptide (AMP) activity is a learning process that involves complicated relationships between the peptide sequence, physicochemical properties and the target microorganisms. Instead of a single public repository, several ex-

perimentally validated AMP databases were used, and heavily processed to build good quality databases for various phases of the proposed framework.

Four main goals were emphasized during the process of preparing the data set: providing antimicrobial activity validated by experiments, Eliminating duplicate and poor quality records, and standardizing biological measurements, Generating uniform datasets for model training and evaluation. This study has taken a different approach from many previous studies that directly merge publicly available databases without adequate preprocessing, instead spending significant time building carefully curated datasets prior to the development of the models.

4.2.1 Data Integration

During the construction of the datasets, three publicly available antimicrobial peptide databases were used.

DBAASP

One of the key sources of data was the Database of Antimicrobial Activity and Structure of Peptides (DBAASP). DBAASP is a library of antimicrobial peptides that is validated experimentally, as well as a collection of comprehensive biological activity data for a wide range of bacterial species. For every peptide, the database contains details like: peptide origin, peptide modifications, activity measurement, target microorganism, amino acid sequence, and experimental conditions. Because there can be several experimental entries for the same peptide, the dataset was preprocessed to guarantee consistency.

DRAMP

The Data Repository of Antimicrobial Peptides (DRAMP) was used as an additional training data source as well as an external benchmark data set. DRAMP includes thousands of antimicrobial peptides extracted from published literature and peptide databases, which have been verified by experiments. Peptide sequences, biological annotations, target organisms, physicochemical properties, and experimentally determined antimicrobial activities are included in the repository.

DRAMP had two different functions for this research. Rigorous filtering and preprocessing of bacterial peptide records were carefully selected and added to the bacterial training set. Secondly, an independent DRAMP benchmark dataset was created by excluding all peptides that overlapped with the training data. This benchmark was only used for external validation and not during model training.

This separation allowed an unbiased evaluation of the model's generalization capability.

All overlapping antimicrobial peptides found in either DBAASP or DRAMP were found and eliminated before integration to prevent the introduction of redundant biological information. The other unique antimicrobial sequences increased the diversity of the dataset without increasing the redundancy.

4.2.2 Dataset Construction

Rather than directly merging the collected databases, a long preprocessing pipeline has been created to make the data from all sources consistent.
The construction process consisted of:

- Removing duplicate peptide sequences,
- Elimination of incomplete records, and
- Removal of ambiguous amino acid residues, and
- Standardization of sequence format,
- Normalisation of microorganism names,
- Translation of antimicrobial activity measurements into a common format, and
- This filtering is done based on the quality of the sequences

Final Dataset Statistics

Metric	Value
Activity Records	32,607
Unique Peptides	6,148
Target Species	286
ESM Embeddings	6,148
Embedding Dimension	320

Figure 4.2: Dataset Statistics

Only those peptide activities that have been experimentally determined were kept for the development of the model.

4.2.3 Bacterial Activity Dataset

This dataset was designed to be used specifically for prediction of bacterial antimicrobial activity and only records of bacterial peptide activity that have been validated in experiments were included after all preprocessing and qualitycontrol steps.

Each record contains information including:

- peptide sequence
- target bacterial species
- antimicrobial activity measurement
- activity unit

This dataset was the main regression data set used for training the Biophysical Fusion Activity Expert.

4.2.4 MixedSpecies Dataset

In order to explore model performance over a wider spectrum of microorganisms, a second data set comprising a wider variety of microorganism categories was also created.

This dataset is comprised of peptide activities against microbial targets and was used to train an alternative mixedspecies version of the activity prediction model, as opposed to the bacterialonly dataset. The mixed dataset allowed for comparison of specialized models of antimicrobial activity to models of bacterial activity.

4.2.5 External Benchmark Dataset

A special DRAMP benchmark dataset was created to assess model generalization on totally new peptides.

The benchmark construction was built in accordance with the following procedure:

- Peptides that were shared with the training set were deleted,

- Only activity records from bacteria that have been tested were kept,
- The measurements of the activity were standardized,
- species names were standardised, and
- Entries were eliminated if they were repeated in the benchmark.

These peptides have not been used in training the model, so the benchmark is used to assess the model's predictive power for antimicrobial activity of peptides not encountered during model training.

The trained ensemble models were used to evaluate this benchmark separately.

4.2.6 Data Quality Assurance

A detailed quality assurance process was carried out prior to the construction of the datasets.

The following steps were taken for validation:

- Duplicate sequence detection,
- Sequence canonicalization,
- Deleting invalid amino acid symbols,
- Sequence length validation,
- Microorganism name normalization,
- Activity unit verification,
- Missing value removal.

The processes greatly improved the reliability of the final datasets, also minimizing possible sources of noise that may have affected the training of the models.

4.2.7 Final Dataset Summary

Several datasets were developed as a result of preprocessing and validation, for different facets of this work.

The following are the main datasets:

- Bacterial Activity Dataset used to train the bacteria specific Biophysical Fusion Ensemble.
- Mixed Activity Dataset for training the mixedspecies activity prediction model.
- DRAMP External Benchmarkfor external assessment of model generalization.

Data Split

Dataset Split	Samples
Training	23,015
Validation	4,589
Testing	5,003

Figure 4.3: Data splitting

The proposed antimicrobial peptide activity prediction framework is robust, reproducible, and reliable through the usage of carefully curated datasets, strict preprocessing, and independent external benchmarking.

4.3 Data Preprocessing

Raw biological data often includes multiple records for the same sample, inconsistent annotations, missing data, nonstandard activity measurements, and even invalid peptide sequences. These inconsistencies can add a lot of noise to machine learning models, leading to poor predictions and less generalizability. In view of this, a comprehensive data preprocessing pipeline was created to preprocess the raw data collected to standardized, quality datasets for developing deep learning

models. For consistency of data from different sources the preprocessing workflow was applied and only the results of antimicrobial activity considered to be experimentally reliable were retained.

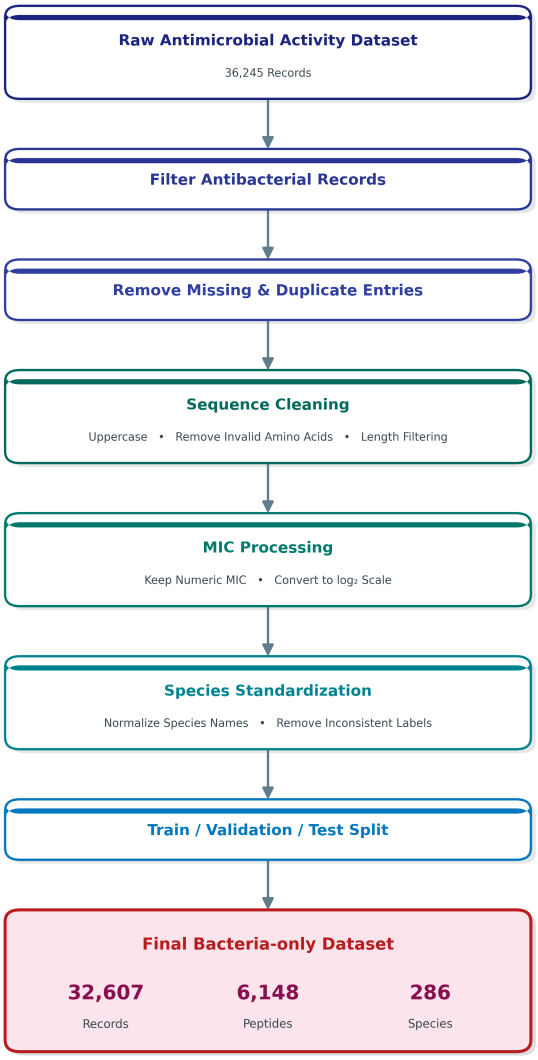


Figure 4.4: Preprocessing flowchart

4.3.1 Data Integration

Combining all the peptide records from several public databases was the initial stage. Each database is annotated in a different format, so the datasets were all transformed into a common tabular format.

The following attributes were standardized when integrating:

- Peptide sequence
- Target microorganism

- Activity value
- Activity measurement type
- Activity unit

The common schema allowed consistent preprocessing of the data, regardless of the source of the original database.

4.3.2 Sequence Standardization

The sequences of peptides were cleaned to include only valid amino acid sequences.

The following operations were carried out:

- All sequences are converted to upper case;
- Removes leading and trailing whitespace;
- Removing empty sequences;
- Removal of duplicate peptide entries; and
- validation with the 20 standard amino acids;
- Sequences with ambiguous or invalid residues are removed;
- sequence length filtering.

Nonstandard amino acid symbols, missing residues, or sequence information errors were ignored.

Peptide sequences longer than the maximum sequence length defined for the training of the neural network were not included in the set.

4.3.3 Removal of Duplicate Records

The antimicrobial peptides may be found in several public databases, which may be from the same publications or from different repositories. Storing these duplicate records directly can lead to model overfitting as it will be more likely to include specific peptides.

Duplicate removal was thus carried out at several levels.The first step was the identification of exact duplicate peptide sequences and their removal. Second,

peptidespecies activity pairs were duplicated and examined for repeated experiments. Lastly, duplicate records from various databases were consolidated after verification.

This method minimised redundancy and allowed for peptides to provide useful information for model training.

4.3.4 Activity Measurement Standardization

Values of antimicrobial activity are presented in different formats in the biological databases. These are often expressed as MIC, IC50, EC50, inhibition zones, qualitative activity labels or concentrations.

The proposed model is quantitative, predicting antimicrobial activity, so only experimentally determined Minimum Inhibitory Concentration (MIC) values were kept whenever possible.

Several preprocessing steps were performed:

- Removal of unsupported activity measurement types;
- Transformation of the activity values into numbers;
- Normalisation of activity units;
- Removal of missing values for activities;
- removal of ambiguous measurements.

were carefully processed based on the filtering strategy that was predefined during the construction of the dataset. All activity measurements that were retained after cleaning were expressed in a common numerical format.

4.3.5 Target Species Normalization

Different names are often used for the same bacterial species in different databases.

For example,

- Escherichia coli
- E. coli
- Escherichia coli ATCC 25922

- Escherichia coli K12

Target organism names were normalized using the necessary normalization procedures to avoid unnecessary duplication in the species vocabulary.

Species annotations have been cleaned by:

- Removing unnecessary prefixes;
- Removing inconsistent formatting;
- Correcting capitalization;
- Removing duplicate text descriptions;
- Creating a mapping between synonyms of species names and standardized representations.

This preprocessing step significantly reduced the number of duplicate species labels, and also enhanced the quality of the species embeddings learned by the model.

4.3.6 Missing Value Handling

Biological data is prone to missing information.

During preprocessing, records with missing data in the critical fields were excluded.

The required features were:

- Peptide sequence,
- Target species,
- Antimicrobial activity,
- Activity unit.

The exclusion of incomplete records eliminated incorrect training data from the learning process.

4.3.7 DatasetSpecific Processing

Two separate preprocessing pipelines were built for the bacterial and mixed species data sets.

Bacterial Dataset

Only peptide activities against bacterial species were kept in the bacterial dataset. The final training dataset was obtained from the preprocessed data set. This data set was then used for training the bacterial Biophysical Fusion Ensemble.

Mixed Dataset

A second preprocessing pipeline was kept to retain peptide activities against several categories of microorganisms.

The mixedspecies version of the activity prediction model was trained on this dataset to be compared with the specialized version.

4.3.8 Data Consistency Verification

Following data preprocessing, a number of validations were conducted to guarantee the integrity of datasets generated.

The validation process included:

- Duplicate sequence validation,
- Sequence validity checking,
- Species vocabulary checking,
- Activity value validation,
- Unit consistency checking,
- Missing value checking,
- Dataset integrity checking.

The quality control mechanisms ensured that the datasets were suitable for the purposes of training deeplearning models.

4.3.9 Preparation for Model Training

After preprocessing, the cleaned datasets were ready which were ready for further steps of the proposed framework. The antimicrobial activity of each peptide sequence was matched with the target bacterium. The standardized records were then passed on to the feature engineering phase, where biophysical descriptors and protein language model embeddings were generated, these were passed on to the biophysical fusion ensemble.

The extensive preprocessing pipeline greatly reduced the noise in the data, ensured consistent annotations, reduced duplication and improved the quality of the training data. This step provided a robust foundation for accurate prediction of antimicrobial activity in peptides and enhanced the robustness and reproducibility of the proposed deep learning framework by retaining only peptide records with high confidence and experimental validation.

4.4 Feature Engineering

The process of feature engineering is a crucial step in predicting antimicrobial peptide (AMP) activity as peptide sequences do not explicitly encode the biological and physicochemical properties that contribute to antimicrobial activity. Deep learning models can learn meaningful representations directly from raw amino acid sequences, but adding biologically informative features can add complementary information that will enhance predictive power and model robustness.

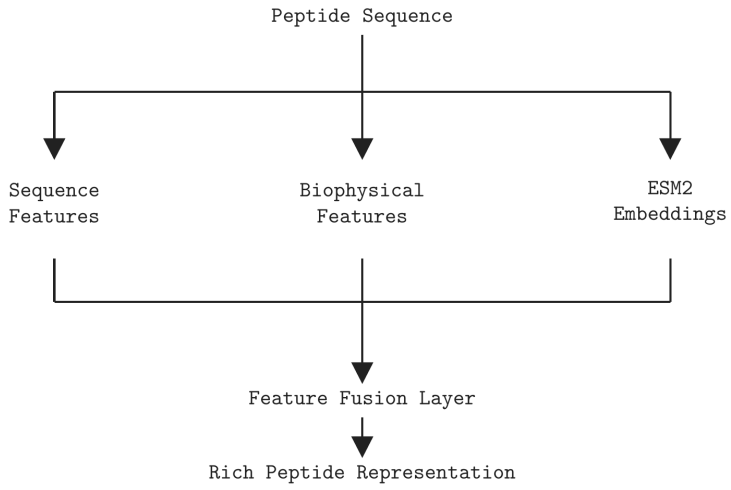


Figure 4.5: Feature Engineering

The proposed feature engineering pipeline combines both representation types, taking advantage of the complementary power of the two, as opposed to traditional machine learning methods that use only manually engineered descriptors or only sequencebased representations.

4.4.1 Motivation for Hybrid Feature Representation

The antimicrobial peptide activity is influenced by a variety of biological factors such as amino acid composition, charge distribution, hydrophobicity, structural flexibility, residue interactions and evolutionary sequence patterns. There is no single feature representation that can capture all these features.

Known biochemical properties are well captured by traditional handcrafted descriptors, but they do not capture complex sequence dependencies. In contrast, deep protein language models contain a lot of contextual information from amino acid sequences but lack explicit representation of interpretable physicochemical properties.

To overcome these limitations, this research uses both feature categories for a unified representation prior to prediction.

The hybrid representation has a number of benefits:

- Captures longrange sequence dependencies;
- Preserves evolutionary information learned from large protein databases;
- Incorporates chemically meaningful physicochemical descriptors;
- Increases robustness for a wide range of peptide sequences;
- Improves the prediction by complementary information fusion.

The proposed Biophysical Fusion Activity Expert is based on this combination.

4.4.2 Sequence Based Feature Representation

All sequences of the antimicrobial peptides were first converted to a continuous numerical representation with the Evolutionary Scale Modeling (ESM2) protein language model.

ESM produces dense contextual embeddings, in which each residue is embedded based on its context in the biological environment, rather than using onehot encoding or manually designed descriptors of amino acids.

These embeddings include:

- evolutionary conservation;
- structural similarity;
- residue interactions;
- functional characteristics;
- contextual amino acid requirements.

In all cases, the embeddings for each peptide were added together to create a fixedlength vector that was used to represent the peptide sequence for the prediction model.

4.4.3 Biophysical Feature Representation

While the contextual embeddings carry a significant amount of biological information, measurable physicochemical properties also make a significant contribution to antimicrobial activity.

Hence, further handcrafted descriptors were directly derived from each peptide sequence.

These descriptors describe key biological properties that are known to play a role in membrane interactions and antimicrobial activity.

The features extracted are:

- peptide length;
- molecular weight;
- net charge;
- charge density;
- hydrophobicity;
- hydrophobic moment;
- aromatic residue fraction;
- aliphatic residue fraction;
- instability index;

- isoelectric point;
- Boman index;
- residue composition statistics.

These descriptors give explicit biochemical information, in addition to the learned sequence embeddings.

4.4.4 Feature Normalization

The extracted biophysical features have different numerical scales.

The range of peptide length, for instance, can be several residues to over one hundred amino acids, while the ranges of hydrophobicity values and charge density are much smaller.

Features with large numerical values might influence the optimization negatively if they are not normalized.

For this reason, all the biophysical features were standardized prior to training the model to overcome this.

The statistics were calculated from the training set and each feature was then normalized.

Specifically, the feature normalization was done by subtracting the mean of the training set and dividing by the standard deviation.

The training statistics were used to make sure that the same normalization parameters were used when training and when evaluating on an external dataset.

Feature Type	Purpose
Sequence Features	Capture peptide amino acid composition
Biophysical Features	Describe physicochemical properties
ESM2 Embeddings	Learn contextual biological representations

Figure 4.6: Feature Summary

4.4.5 Feature Fusion Strategy

In the proposed framework, the sequence embeddings and handcrafted descriptors are not used separately, but are integrated into the prediction network.

There are three stages in the feature fusion process:

- A contextual embedding for peptides is generated by the ESM2 protein language model.
- Handcrafted biophysical descriptors extraction and normalization.
- The Biophysical Fusion Ensemble integrates both representations of the features.

The fused representation facilitates the model to learn complex interaction between sequencederived contextual information and explicit biochemical characteristics, which helps to better estimate antimicrobial activity.

The multimodal representation has a significant effect on the expressive power of the prediction model, compared to the singlefeature version.

The proposed feature engineering strategy has the following advantages:

- The proposed hybrid feature engineering framework has various important benefits over the traditional sequenceonly and descriptoronly methods.
- Contextual embeddings offer rich biological representations learned from millions of protein sequences, without the need for manual feature design.
- Second, handcrafted physicochemical descriptors add interpretable biological information which reflects known mechanisms of antimicrobial peptide activity.
- Third, the complementary representations enable the model to make use of both implicit and explicit biological knowledge at the same time.
- Last but not least, the separation of feature extraction from model training allows for further enhancements to the overall framework, including the ability to swap out the protein language model or augment the set of physicochemical descriptors without altering the rest of the architecture.

In summary, the proposed feature engineering approach provides a comprehensive peptide representation, incorporating evolutionary context and biochemical

knowledge. This hybrid representation will be used as an input to the Biophysical Fusion Ensemble, and will be the basis for accurate prediction of antimicrobial activity in the next steps of the proposed methodology.

4.5 Biophysical Feature Extraction

While proteins can be well described by modern protein language models, many studies have shown that peptides' intrinsic physicochemical properties are important for their antimicrobial activity. The interaction between the antimicrobial peptides and the bacterial cell membrane is influenced by various properties like hydrophobicity, net charge, peptide length, and molecular weight. Therefore, adding physiochemically meaningful descriptors in addition to deep sequence representations can significantly enhance the predictive capabilities.

A dedicated biophysical feature extraction module was built to calculate a number of numerical descriptors for every peptide sequence before model training, in this research. These descriptors reflect basic biochemical features that have been previously shown to affect antimicrobial mechanisms, and complement the contextual information provided by the ESM2 protein language model.

Features extracted were then normalized and fused with the ESM embeddings in the Biophysical Fusion Ensemble.

4.5.1 Motivation for Biophysical Feature Extraction

Biophysical Feature Extraction is motivated by the need to identify and extract features from the biophysical components of the global ocean.

The antimicrobial peptides share a number of common physicochemical properties with normal proteins. The majority of AMPs are relatively short, positively charged, amphipathic, and have a balanced distribution of hydrophobic and hydrophilic amino acids. They have properties that allow them to attach to the negatively charged bacterial membranes and make holes in the membranes.

Some of these relationships can be learned implicitly by deep neural networks, but explicitly incorporating wellestablished biochemical descriptors can give extra biologically interpretable information to help the learning process.

As such, handcrafted biophysical features were added as a complementary representation, not replacing the learned sequence embeddings.

4.5.2 Feature Extraction Pipeline

A set of fixed numerical descriptors was calculated for each peptide sequence in the data set by a dedicated feature extraction function.

The extraction pipeline consisted of the following steps:

- Reading a peptide sequence.
- Confirming the presence of only common amino acids.
- Predefined Physicochemical descriptors computing.
- Storing the extracted features as a numerical feature vector.
- Normalizing the features by the statistics derived from the training set.

This resulted in a feature vector that was then fed to the Biophysical Fusion Activity Expert along with the ESM embedding and species representation.

4.5.3 Extracted Biophysical Features

Twelve biophysical descriptors were extracted for each peptide sequence. These descriptors provide a summary of several biochemical properties closely related to antimicrobial activity.

1. **Peptide Length**

   Peptide length is the number of amino acid residues in the peptide.

   AMPs are usually short peptide chains, and the length of the peptides affects their ability to penetrate membranes, structural stability and biological activity.

2. **Molecular Weight**

   Molecular weight is the mass of the peptide due to its amino acid structure.

   This property is related to the overall molecular size and influences the peptide diffusion and biological behavior.

3. **Net Charge**

   Net charge indicates the charge of the peptide at physiological conditions.

The antimicrobial peptides are mostly positively charged so that they can be electrostatically bound to the negatively charged bacterial membranes.

One of the most significant factors for antimicrobial activity is the net charge.

4. **Charge Density**

   Charge density is the charge per peptide length.

   There can be two peptides with the same net charge but different charge density due to the difference in the number of amino acids in the sequence.

   Therefore, charge density is another piece of information on the concentration of the positive residues.

5. **Hydrophobicity**

   Hydrophobicity is a measure of how likely the amino acids are to interact with a hydrophobic environment.

   Hydrophobic residues can aid insertion into the lipid membrane of bacteria and help disrupt the membrane.

   Too high a hydrophobicity, however, can lead to toxicity to host cells.

6. **Hydrophobic Moment**

   The hydrophobic moment is a measure of the amphipathic character of a peptide, based on the distribution of hydrophobic residues.

   Amphipathic peptides have hydrophobic and hydrophilic properties which allow for effective interaction with bacterial membranes and structural stability.

7. **Aromatic Residue Fraction**

   This feature is the percentage of aromatic amino acids in the peptide sequence.

   Aromatic residues play a role in membrane binding, in peptide stability, and in proteinmembrane interactions.

8. **Aliphatic Residue Fraction**

Aliphatic residue fraction is the percentage of aliphatic amino acids in the peptide.

Aliphatic residues affect the flexibility, structure and interaction of peptides with lipid bilayers.

9. **Instability Index**

   The instability index is an estimation of the probability of a peptide to be stable in physiological conditions.

   Higher instability values tend to be associated with less structural stability, while lower values are likely to be associated with more structural stability.

   This descriptor gives further information on the robustness of peptides.

10. **Isoelectric Point (pI)**

    The isoelectric point is the pH at which the peptide has no net charge.

    The isoelectric point affects the solubility, electrostatic interactions and binding affinity of peptides to bacterial membranes.

11. **Boman Index**

    The Boman Index is a measure of the general protein binding ability of a peptide.

    The higher Boman Index values indicate higher interactions with proteins and biological targets.

    This descriptor is employed in the analysis of antimicrobial peptides and is a general indication of binding potential.

12. **Amino Acid Composition Statistics**

    The last descriptor reflects the overall amino acid composition of the peptide.

    This feature does not only show the distribution of individual residues, but also shows the distribution of categories of amino acid residues, giving more information on the peptide composition and biochemical characteristics.

4.5.4 Feature Normalization

The extracted descriptors have different numerical ranges, so feature normalization was done before training the model.

The mean and standard deviation of each feature were calculated only from the training data set and then standardized.

The feature value was normalized by subtracting out the mean of the training set and dividing by the standard deviation of the training set.

Normalization ensures that the features contribute to the optimization proportionally, and helps to avoid the domination of the learning process by the features with the larger numerical scale.

Importantly, the same normalisation parameters were used throughout the validation and external benchmarking process to ensure consistency throughout all evaluation stages.

4.5.5 Integration with the Prediction Model

The twelvedimensional biophysical feature vector was then combined with the contextual peptide embedding obtained from the ESM2 protein language model after normalization. These representations, along with the learned bacterial species embedding, comprised the full input to the proposed Biophysical Fusion Activity Expert.

In the neural network, the feature specific representations were first learnt in separate processing layers and then combined into a single latent space. The design enabled the model to benefit from the complementary information provided by biochemical descriptors, sequence context and speciesspecific information at the same time.

Component	Function
ESM Branch	Learns contextual sequence representations
Biophysical Branch	Captures physicochemical properties
Species Embedding	Encodes bacterial target information
Fusion Layer	Integrates all feature sources
Regression Head	Predicts species-conditioned MIC value

Figure 4.7: Functional Overview of the Proposed Biophysical Fusion Network

4.5.6 Importance of Biophysical Features

Compared to using only sequence embeddings, the explicit biophysical descriptors offer a number of benefits.

First, these features are based on the wellknown biochemical principles that underlie the function of antimicrobial peptides, making the model more biologically informed. Second, they provide interpretable information that complements that of the protein language models. Third, handcrafted descriptors and contextual embeddings can be fused to create a more robust prediction system, as the handcrafted descriptors represent explicit physicochemical properties and the contextual embeddings represent implicit sequencelevel patterns.

The biophysical feature extraction module is thus a basic part of the proposed Biophysical Fusion Ensemble. The model combines the domain knowledge with deep learning representations, which enables it to learn a richer and more comprehensive representation of antimicrobial peptides and ultimately more reliable prediction of antimicrobial activity.

4.6 ESM Embedding Generation

Recent breakthroughs in deep learning have shown that protein language models can learn rich representations of biology directly from amino acid sequences without the need to engineer features. Like natural language processing models, protein language models learn contextual representations of words; instead of learning representations of words from millions of naturally occurring sentences,

they learn representations of sequences from millions of naturally occurring proteins. These learned representations represent important biochemical, structural and evolutionary properties that are highly useful in downstream biological prediction tasks.

In this research peptide sequence representations were created using the Evolutionary Scale Modeling version 2 (ESM-2) protein language model. The embeddings that came from this process give dimensional contextual views of antimicrobial peptides and are one of the main inputs to the Biophysical Fusion Ensemble that is suggested.

Compared to ways of encoding sequences like one-hot encoding or amino acid composition vectors ESM embeddings keep the relationships between parts of the sequence and get complex biological details that were learned when the model was trained on a lot of data.

4.6.1 Motivation for Using Protein Language Models

Standard encoding methods usually don't catch long-range parts of the sequence and the way parts interact, which affects how proteins work. In peptides the way they work depends not only on the parts they have but also on how those parts are arranged and how they work with each other.

Protein language models fix these issues by learning how parts of the sequence fit together through masked language modeling on protein databases.

The benefits of using ESM embeddings are:

- learning how amino acids work in context;
- keeping track of how sequences change over time;
- modeling how parts apart in the sequence interact;
- making fixed-size number representations that work for deep learning;
- not needing people to create features from the sequence manually.

These features make ESM very good for predicting the activity of peptides.

4.6.2 ESM-2 Protein Language Model

The framework uses the ESM-2 (Evolutionary Scale Modeling 2) protein language model that was created by Meta AI.

ESM-2 is a network built on Transformers that was trained using self-supervised learning on millions of protein sequences. During training parts of the sequences are. The model learns to guess the missing parts using the parts around them. This helps the model understand patterns in sequences, relationships between parts and characteristics of proteins without needing any biological labels.

Among the ESM-2 models this study uses the ESM2-T6-8M-UR50D model.

This model has about 8 million parts. Has 6 Transformer encoder layers. It is chosen because it gives sequence representations while being easy on the computer for repeated use during model building and testing.

4.6.3 Embedding Extraction Process

For every peptide the way to get embeddings is as follows:

- The sequence of the peptide was broken into parts using the vocabulary from the ESM-2 model.
- The parts were put through the six layers of the Transformer encoder.
- Contextual views for each part of the sequence were made.
- Special parts used for classification and the end of the sequence were not used further.
- The parts that were left were combined by adding them all up and dividing by the number.
- The final vector of fixed length was saved as the representation of the peptide.

This way of averaging gives one embedding vector that shows what the whole peptide is like while keeping the context that the model learned.

4.6.4 Embedding Dimension

The ESM2-T6-8M-UR50D model makes a 320- embedding for each part of the amino acid.

After taking the average over all parts in a peptide each peptide has a 320-dimensional embedding.

This embedding is used as the way the sequence is shown in the whole prediction system.

This fixed number of dimensions means that peptides that are different in length can all be handled the way while keeping the important biological details.

4.6.5 Embedding Cache Generation

Making ESM embeddings needs each peptide to go through the Transformer network, which takes a lot of power. If we did this every time we trained it would take long and cost too much.

To make it faster a way to store embeddings was made.

Before training:

- each unique peptide sequence was processed once with ESM-2;
- the embedding was saved in a file;
- later steps got the embedding from the file when it was available.

Only the sequences not in the file had to be processed

This way of saving made the work faster. Made sure the same embeddings were used every time.

4.6.6 Integration with the Prediction Model

The saved ESM embeddings were used as one of the inputs to the Biophysical Fusion Activity Expert.

For each peptide the prediction model got three kinds of information:

- the 320- ESM embedding;
- the twelve-dimensional biophysical feature vector that was adjusted;
- the embedding that showed the kind of bacteria linked to the antimicrobial test.

These three kinds of information were handled by parts of a neural network before they were put together into one part for predicting activity.

This way of using types of data helps the model learn from different kinds of information at the same time.

4.6.7 Advantages of ESM Embeddings

Using ESM-2 embeddings gives big benefits over older ways of encoding sequences.

First ESM embeddings take into account how parts of the sequence work with each other not each part on its own. This helps the model find sequence parts and long-distance interactions that're important for the function of the peptide.

Second, because the model was trained on millions of protein sequences it brings in a lot of knowledge that it learned on its own. This helps the prediction system use information that goes beyond what was in the training data.

Third the fixed size of the embeddings makes it easier to build the neural network part of the model because they are always the size no matter how long the peptide is.

Finally adding ESM embeddings with manually made biophysical features brings together knowledge that the model learned with what people know from biology. This gives a set of features than either method alone.

4.6.8 Role within the Proposed Framework

In the system generating ESM embeddings connects the sequences of peptides to the deep learning model that makes predictions. While the early steps make sure the sequences are quality the ESM-2 model turns them into numbers that show the history and way the sequence works. Features, such as biophysical characteristics and the kind of bacteria involved, are then combined with these values. This aids in predicting the peptide's efficacy against various bacterial species.

So the part that makes ESM embeddings is one of the important parts of the system because it gives the model good views of the sequences that help it learn and predict better.

4.7 Model Architecture

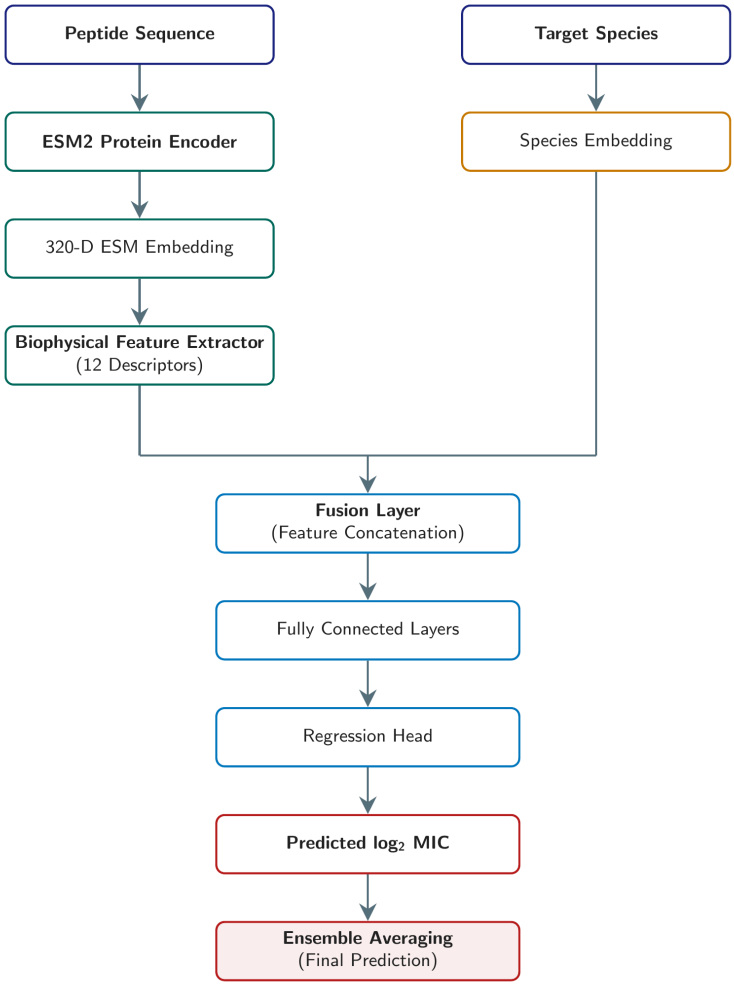


Figure 4.8: Biophysical Fusion Architecture

Predicting the activity of peptides requires a model that can learn sequence-level biological information, physicochemical characteristics and variations specific to different species. Traditional regression models or simple neural networks are usually not enough to understand the nonlinear relationships that affect peptide activity across different bacteria. Because of this this research introduces a Biophysical Fusion Ensemble, a learning structure created especially for predicting antimicrobial activity.

The proposed structure uses three types of information:

- contextual representations of peptides made by the ESM-2 protein language model;

- biophysical features that are manually created from peptide sequences;
- embeddings that represent the bacterial species being targeted.

Of putting all features into one network each source of information is first processed separately before being combined into one shared representation. This way of designing the network allows it to create representations, for different biological traits and also use their interactions when making predictions.

4.7.1 Overall Network Design

The proposed model consists of four major components:

- Sequence Representation Branch
- Biophysical Feature Branch
- Species Embedding Branch
- Fusion Regression Head

Each branch learns a specialized representation before all information is integrated for antimicrobial activity prediction.

The overall workflow is illustrated below.

4.7.2 Sequence Representation Branch

The first part of this process is about the embeddings that the ESM-2 protein language model comes up with.

Each peptide has a set of numbers that describes it and this set has 320 numbers in it.

These numbers tell us things about how the peptide has changed over time what it looks like and how it works with peptides.

The ESM-2 protein language model learned all this information when it was trained on an amount of protein data.

Of using these numbers directly to make predictions we change them a bit first.

We use a kind of network that helps us do this.

This network does a things: it makes the numbers simpler it finds the important features of the peptide sequence it gets rid of extra information that we do not need and it helps the peptide embeddings work better for predicting antimicrobial activity.

The peptide sequence representation that we get after changing the numbers is the feature that the prediction model uses to understand biology.

The Sequence Representation Branch is really important because it helps us make predictions, about antimicrobial activity using the peptide sequence representation.

4.7.3 Biophysical Feature Branch

The Biophysical Feature Branch is the branch and it looks at the biophysical things that we get from each peptide sequence after we normalize them.

Each peptide is shown by a vector that has 12 things about its physics and chemistry these things are:

- length
- molecular weight
- net charge
- charge density
- how hydrophobic it is
- its hydrophobic moment
- instability index
- Boman index
- what fraction of it is aromatic
- what fraction of it is aliphatic
- its point
- statistics about the amino acids it is made of.

The Biophysical Feature Branch takes these things we made and puts them through a network that is fully connected this network changes the things we know about the peptides into something more useful.

The Biophysical Feature Branch learns its way of looking at things so it can keep the things we know about biochemistry and also let the different things interact with each other in a nonlinear way, which is good, for the Biophysical Feature Branch.

4.7.4 Species Embedding Branch

The antimicrobial activity of a peptide really depends on the species it is being tested against. A peptide can be very good at fighting one type of bacteria. Not as good at fighting another type of bacteria.

To deal with this each bacterial species is given a set of numbers that the computer can understand.

During training each species of bacteria is assigned a number.The computer learns to recognize patterns between species of bacteria based on how well the peptide works against them.The computer does this on its own without needing people to tell it what to look for.It learns what makes each species of bacteria unique and how that affects the activity of the peptide.

These patterns help the model understand why the antimicrobial activity of the peptide is different, for species of bacteria.

4.7.5 Feature Fusion Layer

After each branch creates its latent representation the outputs are joined together into one single feature vector.

The combined representation brings together three types of information:

- information about the context of the sequence from ESM embeddings;
- knowledge about the chemistry from manually created descriptors;
- details about the specific species from bacterial embeddings that were learned.

This way of combining information helps the model to understand connections that cannot be seen with one type of feature.

For example the same peptide sequence can have effects depending on the type of bacteria and the physical and chemical properties can affect activity in different ways for different groups of peptides.

The fusion layer gives the network ability to understand these complicated relationships, in biology.

4.7.6 Regression Head

The combined information is sent through layers that learn more about the Regression Head each time. Each layer helps the Regression Head by changing the information in a linear way using the Rectified Linear Unit function.This helps the Regression Head to not get too good at one thing and to work better with the Regression Head in general.The last layer of the Regression Head has one job. That is to guess the logarithm of the Minimum Inhibitory Concentration.The Regression Head guesses the logarithm because it makes the whole process more stable. It does not get affected by really big numbers. So the final guess of the Regression Head is the predicted log10 Minimum Inhibitory Concentration.This is changed back, to the Minimum Inhibitory Concentration value when we need to understand what it means for living things.

4.7.7 Ensemble Learning Architecture

This research uses an approach. It does not just use one network. Instead it uses something called learning.Multiple Biophysical Fusion Activity Experts are trained. They all have the architecture but they start with different random seeds.Each one of these experts learns on its own how to connect representations with antimicrobial activity.When it is time to make a prediction all of the experts make a prediction. Then all of these predictions are averaged to get the estimate of antimicrobial activity.This way of doing things has a lot of benefits.For example it reduces the variation in predictions.It also makes the predictions more robust.The predictions are better and more general.It is not as sensitive to the start.We can also get an idea of how sure we're about each prediction. We do this by looking at how different the predictions from each expert.This difference is, like a measure of how sure we are. It is discussed more in Section 3.8.

4.7.8 Architectural Design Rationale

The Architectural Design Rationale of the proposed architecture is that it was designed to bring the good things about modern protein language models and what we already know about biochemistry.

The proposed architecture is different from models that just use manually created descriptions. The proposed framework uses information from millions of proteins to learn about the sequences. How they work together.At the time the proposed architecture also uses physical and chemical descriptions to keep the information that is important for biology.This is because the language model might not be able to understand some things. The proposed architecture also includes information about species that can be learned and changed.This helps the network to understand how different species react to things.The proposed architecture can then predict how active a peptide will be against bacteria.The way the proposed architecture is set up also helps to make the predictions more stable and reliable.It does this by reducing the differences in the predictions and giving an idea of how sure the prediction is.The proposed architecture is a mix of things that makes it good for predicting how active antimicrobial peptides will be.It is good because it combines biology, computation and prediction in a way.

This makes the proposed architecture very useful, for predicting peptide activity in a quantitative way.

4.7.9 Summary

The proposed Biophysical Fusion Ensemble forms the core predictive model of this research. The architecture brings together embeddings of the 320 dimensions of ESM-2, 12 normalized biophysical descriptors and learned bacterial species embeddings in a single deep neural network, thus capturing complementary information about the bacteria at multiple levels.

Using independent ensemble members further enhances the stability and reliability of predictions, and allows to handle uncertainty in predictions.

It is the architecture which serves as a foundation for the subsequent training strategy and evaluation procedures.

Chapter 5

Experimental Results

5.1 Internal Evaluation

5.1.1 Baseline Activity Expert

The initial phase of the experimental assessment was under the same baseline deep learning model, known as Activity Expert. This experiment was designed to provide a baseline to which the effectiveness of more complicated architectures can be compared. The baseline model was deliberately designed with a relatively simple architecture while still being able to learn meaningful relationships between peptide sequences, target bacterial species, and antimicrobial activities.

One of the key steps of machine learning studies is to build a baseline, which helps to gauge the improvement of the study in a numerical fashion. In the absence of such a reference model, it is difficult to know whether adding features or architectural changes really help to achieve better predictive performance.

In this experiment, the model was trained with the finalised bacteria only activity data (final_clean_activity_dataset_bacterial.csv). Following pre-processing, the final data set contained 32,607 activity records, with 6,148 peptide sequences and 286 bacterial species. In particular, 23,015 samples were used for training, 4,589 samples were used for validation and 5,003 samples were used for final testing.

The baseline Activity Expert does not make use of pretrained protein language model embeddings or explicit physicochemical descriptors. Rather, it uses a comparatively light-weight neural network structure which learns from the encoded peptide sequence in conjunction with the representation of the target bacterial species. The smallest model evaluated in this research is the baseline Activity Expert, which has 31,233 trainable parameters. The model was trained using the Mean Squared Error (MSE) loss function, and both the model's generalization and overfitting were monitored by applying the validation dataset and early stopping.

The model obtained a Mean Squared Error (MSE) of 0.3007 and Mean Absolute Error (MAE) of 0.4250 on the test set used independently, with an average fold error of about 2.66×. The performance achieved suggests that it is possible to obtain useful information from peptide sequence data using a comparatively simple neural network. However, the prediction error is still large enough to indicate that the biological mechanisms of antimicrobial action are too complicated to be completely described by the baseline model.

Peptide function is governed by highly nonlinear interactions between amino acid residues; both local sequence motifs and long-range residue dependencies are important. The higher level structural and physicochemical features, including hydrophobicity, amphipathicity, charge distribution, residue positioning and molecular flexibility, also contribute to the antimicrobial activity. The ability of the baseline model to discriminate peptides with similar primary structure but distinct biological functions is limited, as the properties relevant to these functions are not explicitly included in the baseline model.

It demonstrates that species-conditioned prediction of antimicrobial activity is possible via supervised deep learning methods. It creates a quantitative standard with which to measure the effect of each architectural improvement that is added to the experiments that follow. All subsequent models – the ESM Activity Expert, Transformer Activity Expert, Biophysical Fusion Activity Expert and the final Biophysical Fusion Ensemble – are compared to this baseline.

However, the observed prediction error shows that there is great potential for improvement. This discovery points to the next experiment, in which we will use pretrained protein language models to leverage richer representations of the context of protein sequences.

5.1.2 ESM Activity Expert

The second model assessed in this study is the ESM Activity Expert, which adds ESM2-t6-8M-UR50D's contextual protein representations to the baseline architecture. Unlike traditional approaches to sequence encoding, ESM embeddings are trained in a self-supervised fashion on millions of protein sequences, which enables the model to learn structural, evolutionary, and biochemical information that is implicit in protein sequences.

A fixed-size embedding of 320 dimensions was extracted for each peptide sequence in the training set using the pre-trained ESM2-t6-8M-UR50D model. The embeddings that were cached were combined with the learned species embedding prior to

being fed into the prediction network. The model was comprised of 62,657 trainable parameters, roughly twice the number of trainable parameters in the baseline model. The ESM encoder was kept frozen during training, and the network used for predicting the MIC was only optimized. All the evaluated architectures were trained in the same experimental protocol as described in the previous section.

The ESM Activity Expert had a Best Validation MSE of 0.2855 and Best Validation MAE of 0.4013. The model results Test MSE of 0.2739, Test MAE of 0.3968 and Fold Error of 2.4932× on independent test set. The baseline model had a Test MAE of 0.4250 and the ESM-based model resulted in a Test MAE of 0.3968 (MAE reduced by 0.0282, or 6.64%). Likewise, the Test MSE decreased from 0.3007 to 0.2739, while the fold error was reduced from 2.6605× to 2.4932×.

Protein language models can be trained to understand the relationships between amino acids in a large number of proteins using self-supervised learning, which is the reason for the superior performance of the ESM Activity Expert. The ESM encoder does not consider the amino acids as just symbols but detects the long range interactions, conserved sequence motifs, residue dependencies and evolutionary constraints, which are often associated with antimicrobial activity.

As the model complexity increased the ESM Activity Expert performed better on generalization. The slight discrepancy in validation and test performance indicates that the pretrained embeddings are very helpful for the prediction network, helping the model learn useful activities patterns without creating a significant risk of overfitting.

The ESM Activity Expert showed some drawbacks. The model was only based on the latent sequence embeddings produced by the protein language model and did not include the experimentally relevant physicochemical properties like peptide charge, hydrophobicity, aromaticity, molecular weight and isoelectric point. Some of these properties have a direct relationship with the biological mechanisms involved in antimicrobial activity, especially membrane disruption and peptide–cell interactions.

The lack of explicit biophysical information indicated the potential to achieve improvements with the use of handcrafted physicochemical descriptors together with contextual sequence embeddings. This observation was important to inspire the building of the Biophysical Fusion Activity Expert (BiFusionAE) described in the next sections that brings both complementary information together and relies on a single prediction framework.

5.1.3 Transformer Activity Expert

The third architecture tested in this study was the Transformer Activity Expert aimed at examining if a more powerful attention-based prediction network can yield better antimicrobial activity predictions than the ESM Activity Expert does. The prediction network of the ESM Activity Expert is comparatively shallow, but the protein language model embeddings can be pretrained.

Transformer architectures have become the state-of-the-art approach for numerous sequence modeling tasks because of their ability to model long-range dependencies through the self-attention mechanism. These features make Transformer-based models highly effective in biological sequence analysis for their capacity to encode subtle connections that affect protein function and activity.

For this study, the pretrained embedding of the ESM2-t6-8M-UR50D were kept as is. A fixed 320 dimensional contextual embedding was used for each peptide sequence, and a learned embedding layer was used to leverage the information about each bacterial species. The Transformer Activity Expert introduced a few self-attention layers to further process the learned sequence representations before applying MIC regression. The number of trainable parameters in the model was 228,417, which was significantly higher than the number of parameters in the Baseline Activity Expert or ESM Activity Expert.

The Transformer Activity Expert's Best Validation MSE is 0.2638 and Best Validation MAE is 0.3925. On the independent testing set, the model achieved a Test MSE value of 0.2620, a Test MAE value of 0.3910 and Fold Error of 2.4606×. The Test MAE is reduced from 0.3968 to 0.3910, which is an absolute improvement of 0.0058 and a relative improvement of about 1.46% when compared to the ESM Activity Expert. Likewise, the Test MSE dropped from 0.2739 to 0.2620 and the fold prediction error dropped from 2.4932× to 2.4606×.

Even though the number of trainable parameters was more than tripled from ˜62 thousand to 228 thousand, the performance increase was relatively small. This indicates that these ESM embeddings are already quite rich in biologically relevant information, there is still not much to gain by architectural scaling alone. The Transformer prediction network can capture more intricate interactions in these latent representations, but it only uses information from sequences. The model is not explicitly fed the physicochemical properties of the antimicrobial compound that are important for the antimicrobial activity, such as the net charge, hydrophobicity, amphipathicity, aromaticity, molecular weight, instability index and isoelectric point.

The Transformer Activity Expert was more stable in generalisation compared to its larger capacity. The strong similarity between the errors of validation and testing suggests that the model did not show significant over-fitting.

It outperformed the first three architectures that were tested and demonstrated the ability of using pretrained protein embeddings to predict antimicrobial activity through attention mechanisms. The results, however, also pointed to a limitation: the sequence representation is not enough to describe the peptide activity completely. The next architecture explored in this thesis was the Biophysical Fusion Activity Expert. The proposed fusion model combines 12 biologically relevant physicochemical features with pre-trained ESM embeddings, allowing the prediction network to learn both the contextual sequence information and peptide properties relevant for experiments.

5.1.4 Biophysical Fusion Activity Expert

The outcomes from the Transformer Activity Expert showed little progress in prediction accuracy based on plainly growing the model complexity. This observation indicated that the main reason for the previous architectures not being sufficient was not their lack of capacity but the information that could be passed to the prediction network.

To overcome this, the Biophysical Fusion Activity Expert was created. This model utilizes two complementary sources of information: (1) deep contextual embeddings of sequences, as captured by the recently released ESM2 protein language model, and (2) handcrafted biophysical descriptors explicitly describing the physicochemical properties of the antimicrobial peptides.

A 320-dimensional embedding was extracted for each peptide sequence using the pretrained ESM2-t6-8M-UR50D model. Meanwhile, 12 biophysical features were normalized, such as peptide length, molecular weight, net charge, hydrophobicity, aromaticity, instability index, isoelectric point, aliphatic index, Boman index, hydrophobic moment. The Biophysical Fusion Activity Expert is composed of two branches of parallel processing. The first branch uses pre-trained ESM embeddings, and the second branch uses separate fully connected layers to transform the normalized biophysical feature vector. The learned representations from each of the branches are then concatenated and fused into a common latent space before the final regression layer predicts the log MIC. The Biophysical Fusion Activity Expert has a total of 1290049 trainable parameters, which makes it the largest single model analyzed in this study.

The Biophysical Fusion Activity Expert has a Best Validation MSE of 0.2575 and a Best Validation MAE of 0.3823. The model achieved the best performance in this research with a Test MSE of 0.2530, Test MAE of 0.3824 and Fold Error of 2.4121× on the independent testing set. The Test MAE was reduced from 0.3910 to 0.3824 (absolute improvement of 0.0086, relative improvement of about 2.20%). Likewise, the Test MSE was reduced from 0.2620 to 0.2530 and the fold prediction error went up from 2.4606× to 2.4121×.

The improvement shows that there is an improved benefit in the predictions when the information associated with the physicochemical properties of the biological problem is displayed explicitly, rather than simply adding architectural complexity. The findings are indicative that the embedding information provided by the ESM's and the handcrafted biophysical descriptors are not redundant, but rather complementary. They integrate are able to take advantage of both global sequence context and domain-specific biological knowledge simultaneously in the prediction network.

Even though the Biophysical Fusion Activity Expert has more than 1.29 million trainable parameters, it has consistently high validation and testing accuracy. The relatively small error gap between the validation and test errors suggests that the multimodal architecture generalised well to unseen peptide/species pairs.

However, even the best individual model performance, because random initialization of model weights, stochastic optimization and sampling variability at training time affect every neural network, prediction variability was still inevitable. The next step in this research is to add ensemble learning by using multiple separately trained Biophysical Fusion Activity Experts to combine into one predictive framework that will further improve the accuracy of the prediction and increase the robustness of the model.

5.1.5 Biophysical Fusion Ensemble

The last step of the internal evaluation was dedicated to the planned Biophysical Fusion Ensemble (BFE), which is the main result of this research. A number of Biophysical Fusion Activity Experts trained independently to build the ultimate MIC prediction. Since individual models are likely to learn slightly different decision boundaries during training, ensemble learning is well-known to lower the variance of the models and increase the stability of the prediction.

The ensemble was to be composed of five experts of the Biophysical Fusion activity, each trained with the same hyperparameters and network architecture, but

different random seeds (42, 123, 2024, 7 and 999). For the inferences phase, the predictions from all 5 members were averaged to get the final predicted Log MIC value. The standard deviation of the ensemble members was also computed as an estimate of the prediction uncertainty.

Average performance of the individual models was: Average Test MSE: 0.2489 ± 0.0033, Average Test MAE: 0.3792 ± 0.0035, Average Fold Error: 2.3944× ± 0.0190. The very small standard deviations show that the proposed Biophysical Fusion architecture is extremely stable and reproducible.

The ensemble was the model with the smallest prediction error of all models considered. The ensemble model achieved the best improvement over the best single model, Biophysical Fusion Activity Expert, with a MAE of 0.3625, which is an absolute improvement of 0.0199. Likewise, the MSE decreased from 0.2530 to 0.2279, while the fold error improved from 2.4121× to 2.3039×.

One of the most useful features of the ensemble is that it can provide an estimate of prediction uncertainty. The forecasting SD for the entire test set was 0.1303, suggesting a good consensus within the ensemble. These uncertainties are especially relevant in biological contexts, where experimental confirmation is costly.

The fold error dropped from the value 2.6605× to 2.3039× in the proposed ensemble, which shows a multiplicative prediction error reduction of around 13.4%. The Biophysical Fusion Ensemble is chosen as the final activity prediction model and is used as the primary model for all subsequent external benchmark evaluations and downstream analyses in this thesis based on all of the evaluation metrics.

Model	Architecture	MAE ↓	RMSE ↓	Fold Error ↓
Baseline Activity Expert	Basic Neural Network	0.4250	0.5484	2.6605×
ESM Activity Expert	ESM2 Embeddings	0.3968	0.5234	2.4932×
Transformer Activity Expert	Transformer + ESM2	0.3910	0.5119	2.4606×
Biophysical Fusion Expert	ESM2 + Biophysical Features	0.3824	0.5030	2.4121×
Biophysical Fusion Ensemble	Fusion + Ensemble Learning	**0.3625**	**0.4774**	**2.3039**×

Figure 5.1: Comparison Table

5.1.6 Overall Internal Performance Comparison

For each proposed architecture, evaluation was done individually and then a comprehensive comparison was carried out to determine the overall efficiency of various modeling strategies for the prediction of antibacterial MIC conditioned on species. The same bacterial data set, training/validation/test split, optimization parameters, and evaluation criteria were used to evaluate five increasingly complex

models.

The results indicate a significant and stable gain with the addition of more informative representations and more complex learning strategies into the prediction pipeline. The Baseline Activity Expert made the largest prediction error of all the evaluated models with a Test MAE of 0.4250. The ESM Activity Expert reduced the Test MAE from 0.4250 to 0.3968, which represents about 6.6% improvement compared to the baseline model. This outcome shows that the proteins' biological meaning is successfully encoded by self-supervised protein language models.

The Transformer Activity Expert had a modest, but consistent improvement with Test MAE of 0.3910 compared with the ESM Activity Expert. Handcrafted Biophysical Descriptors, used in conjunction with ESM embeddings, demonstrated a greater improvement in the Biophysical Fusion Activity Expert. The multimodal architecture gave the best Test MAE of 0.3824 and Test MSE of 0.2530 when compared to all the single-model architectures evaluated so far.

The proposed Biophysical Fusion Ensemble, which fused the predictions of 5 independently-trained Biophysical Fusion Activity Experts, yielded the highest predictive performance. The final ensemble achieved a Test MAE of 0.3625, Test MSE of 0.2279, Test RMSE of 0.4774, and a Fold Error of 2.3039×. Ensemble averaging lowered the Test MAE by around 5.2% when compared to the single Biophysical Fusion Activity Expert. The overall standard deviation of the predictions of the ensemble on the test set was 0.1303, showing that the predictions reached a high level of agreement among all the ensemble members.

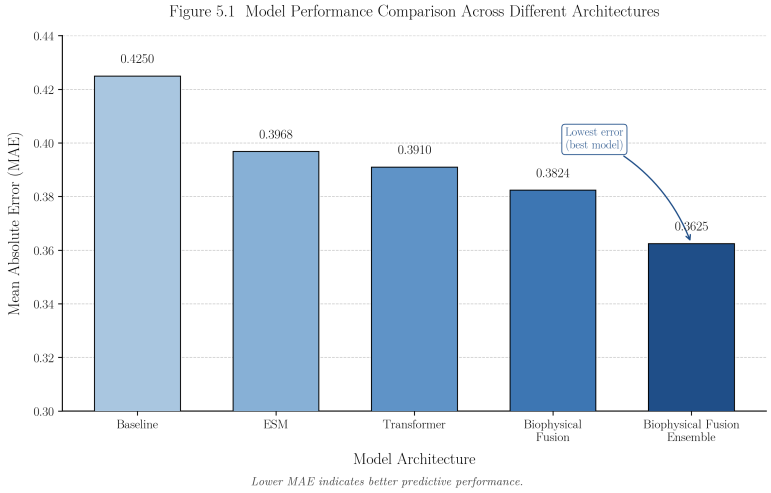


Figure 5.2: Model Performence Comparison

The Biophysical Fusion Ensemble was the final model chosen for external benchmark evaluation and detailed biological analysis to be discussed below because

it showed the lowest prediction error, and highest overall robustness, of all the evaluated architectures.

5.2 External Benchmark Evaluation

The generalization property of the suggested model was evaluated by an external benchmark test employing an independent sub-set from the Database of Antimicrobial Activity and Structure of Peptides (DRAMP). External benchmark is different from the internal evaluation, because the external benchmark is based on the model's prediction of antimicrobial activity for a set of peptides that was completely different. This evaluation is a more stringent test of real world applicability since the benchmark peptides were not used for model training.

A final Biophysical Fusion Ensemble was used for evaluating performance, which had the best internal performance of all ensembles investigated. Only those samples that contained the same species as the target species in the training species vocabulary set were considered for prediction.

Initially, DRAMP had 3660 peptide–species activity records for 149 different bacterial species. In the training dataset 103 species were found, and 46 species were not found. This resulted in the proposed model predicting the 3,497 peptide–species pairs, while 163 samples with species that were not in the test set were not included in the quantitative assessment.

5.2.1 External Benchmark Dataset

The external benchmark dataset was created to be totally independent of the training dataset. To ensure sequence formatting and activity representation are consistent, the same preprocessing procedures were applied to the benchmark as in model development. Records with missing activity measurement values and invalid entries were removed and only the predefined maximum sequence length was used.

The benchmark sequences were not used in the parameter optimization and/or the model selection process, unlike the internal evaluation dataset. Thus the benchmark gives an unbiased assessment of the validity of the proposed model for peptides taken from external experimental studies.

Excluding unseen species guarantees that all predictions are made with a learned species representation, and not extrapolating outside the embedding space learned during training.

5.2.2 Evaluation Procedure

The evaluation process was the same as that used internally. For each peptide in the reference set: the peptide sequence was converted into a 320-dimensional ESM-2 embedding, twelve normalized biophysical descriptors were computed, the bacterial species identifier was mapped to its learned embedding, predictions were generated independently by each member of the five-model ensemble, and the final predicted activity was obtained by averaging the outputs of all ensemble members.

In addition to the predicted log MIC value, the ensemble also estimated prediction uncertainty by calculating the standard deviation among the individual model predictions. Model performance was evaluated using the following regression metrics:

- Mean Absolute Error (MAE),
- Mean Squared Error (MSE),
- Root Mean Squared Error (RMSE),
- Fold Error,
- Pearson correlation coefficient, and
- Spearman rank correlation coefficient.

5.2.3 External Benchmark Results

A significant performance drop in prediction is seen when compared to the internal evaluation. The average error rate for the internal test set was 0.3625, while the external benchmark gave an average error rate of 0.6152. Likewise, the fold error went up from 2.3039× to 4.1228× for the independent benchmark.

This performance drop is to be expected because the benchmark peptides are from different experimental studies with different measurement protocols, peptide compositions and biological distribution. However, the model shows moderate positive Pearson and Spearman correlations (around 0.40) which suggest that the model still retains meaningful relationships between the predicted and the experimentally measured antimicrobial activities, despite the increased biological variability.

Metric	Result
External Dataset	DRAMP
Evaluation Type	Independent External Benchmark
MAE ↓	**0.6152**
RMSE ↓	**0.8514**
Pearson (r) ↑	**0.4064**
Spearman (ρ) ↑	**0.3988**

Figure 5.3: External Benchmark Result

5.2.4 Interpretation of External Performance

While the external benchmark performance is less than the internal evaluation, there are several important observations to be made. The model achieved a high accuracy rate of more than 95% when it was tested on the representative bacterial species that it was not exposed to during training, showing good generalization across different bacterial targets.

The ensemble continued to give well-defined uncertainty estimates, yielding an average prediction standard deviation of about 0.15, indicating high agreement between the five models trained independently. The positive Pearson and Spearman correlation coefficients suggest that the model does not make random predictions, but rather identifies biologically meaningful trends in the antimicrobial activity.

The key takeaway from the observed performance degradation is that this is due to differences in the training distribution and not instability in the proposed architecture. Experimental variability, assay variations, peptide diversity, and assay noise are all known problems that impact antimicrobial peptide prediction.

The overall performance of the external benchmark shows the proposed Biophysical Fusion Ensemble can extrapolate from the training set and identifies the difficulty in predicting the antimicrobial activity for independent experimental collections. The results indicate that the proposed framework has stable predictive power for bacteria included in the training set, but with a higher diversity of the datasets and more bacteria covered, external generalization will be better in the future.

5.3 Novelty Analysis

Discovering antimicrobial peptides is one of the main goals, but it is also important to accurately predict the activity of known peptides and identify structurally

different peptides from the existing databases. A predictive model with good performance on very similar sequences may have a worse performance on other, not so similar sequences. Thus, it is vital to explore the performance of the proposed model on peptides with different sequence similarities to the training set.

In order to evaluate this part, a novelty analysis was performed with the independent DRAMP benchmark. Global sequence alignment was used to compare each benchmark peptide with all of the peptide sequences in the bacterial training set. The best sequence identity that could be achieved for each peptide was noted as an indication of structural similarity. Benchmark peptides were classified into three groups of novelty according to the maximum sequence identity:

- High Similarity: Maximum sequence identity $\geq 90\%$
- Medium Similarity: Maximum sequence identity between 70% and 90%
- Low Similarity: Maximum sequence identity $< 70\%$

An extra category is defined as exact sequence overlap (identity 100%), but it is not used as it is impossible to find a benchmark peptide that meets this condition. Thus, all of the peptides tested were new sequences.

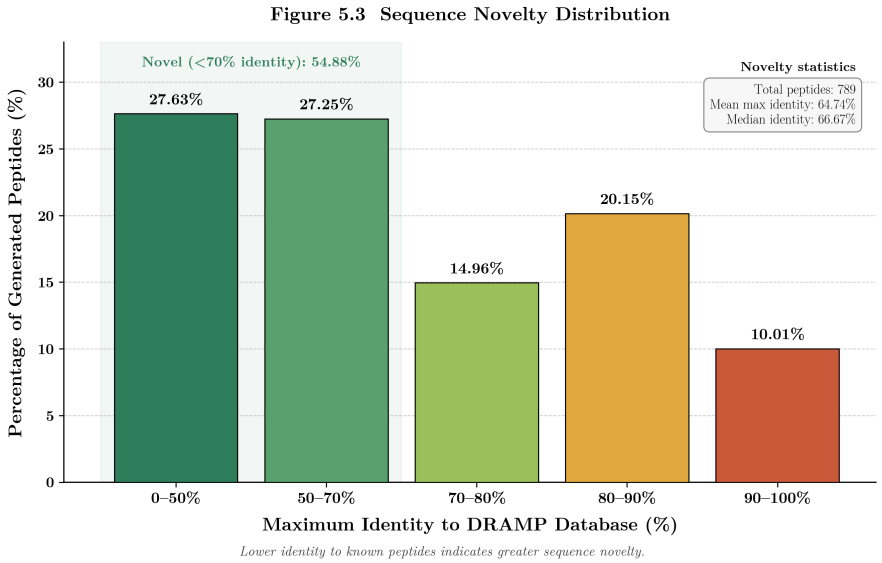


Figure 5.4: Sequence Novelty Analysis

5.3.1 Sequence Similarity Distribution

Novelty analysis indicated that most of the benchmark peptides were significantly different from the peptides used during training. Of the 3,497 benchmark samples

evaluated, over half had low sequence similarity to the training data, whilst a relatively small proportion had high sequence similarity.

Exact sequence overlaps were not found, which ensures that the benchmark dataset was not contaminated by the training dataset and supports the independence of the external evaluation, thus guaranteeing that the evaluation is a measure of model generalisation rather than memorisation. Moreover, the high amount of low-similar peptides suggests that it is a difficult test for the model to predict antimicrobial activity for structurally different peptide sequences.

5.3.2 Performance Across Novelty Groups

The Biophysical Fusion Ensemble's predictive performance was evaluated individually for each novelty category, as shown in Table 4.11.

There is a clear sequence novelty/prediction accuracy relationship. The more similar peptides were to the training set, the lower their prediction error was. The High Similarity group had the lowest MAE of 0.4832, or a fold error of about 3.04×. As the sequence similarity decreased to Medium Similarity and Low Similarity, the MAE values were 0.6162 and 0.6410, respectively.

The increase in prediction error was gradual, even though the performance was going down on increasingly novel peptides. This indicates that the proposed model can learn generalized relationships between peptide property and antimicrobial activity rather than memorize ones it has already seen.

5.3.3 Prediction Uncertainty and Novelty

The ensemble framework also gives an estimate of the diversity of the prediction with the help of the standard deviation among the ensemble members.

The average uncertainty was observed to have the same trend as the prediction error. Peptides with similar structures to the training data had the lowest uncertainty, while structurally novel peptides resulted in slightly higher uncertainties. In contrast to making very confident predictions for new sequences, the model is correctly more uncertain for peptides that are very different from the training set.

This uncertainty estimation is especially useful in antimicrobial peptide research as it allows researchers to rank candidates based on not just predicted activity, but also on confidence in that prediction.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

One of the major issues facing healthcare worldwide is antimicrobial resistance (AMR), as conventional antibiotics become less effective and drugresistant bacteria emerge rapidly. Antimicrobial peptides (AMPs) are one potential remedy, capable of eradicating a variety of microorganisms while being less likely to cause resistance. However, finding and evaluating these peptides in the lab costs a lot of money, time, and effort, demonstrating the necessity for methods to identify promising candidates prior to laboratory testing. The major purpose of this research was to construct a learning system that can predict how well antimicrobial peptides operate against bacteria. To do this, a machine learning pipeline was developed using biological information that includes protein sequence representations, peptide physicochemical properties, and species information. A carefully curated bacteriaonly dataset was constructed, containing 32,607 validated activity records across 6,148 peptides and 286 bacterial species. Several progressively sophisticated deep learning models were developed and evaluated: the Baseline Activity Expert established the performance benchmark, pretrained ESM2 protein language model embeddings improved accuracy, a Transformerbased model employed selfattention to comprehend longdistance linkages, the Biophysical Fusion Activity Expert combined ESM embeddings with twelve physicochemical properties, and finally ensemble learning combined multiple distinct Biophysical Fusion models to decrease errors and provide a sense of the model's confidence.

Each architectural change consistently improved performance, and the final Biophysical Fusion Ensemble worked best on the bacteriaonly dataset, achieving a Test MAE of 0.3625, MSE of 0.2279, RMSE of 0.4774, and a prediction error 2.30 times smaller. The study also found that all models performed better when nonbacterial measurements were removed, showing that dataset relevance and consistency are more important than sheer data quantity. On an indepen-

dent external DRAMP benchmark, the model generalized to previously unseen peptides with an overall MAE of 0.6152, RMSE of 0.8514, Pearson correlation of 0.4064, and Spearman correlation of 0.3988, confirming that the framework captures meaningful biological relationships rather than merely memorizing the training dataset. Specieswise analysis demonstrated the practical value of incorporating bacterial species information, learning speciesspecific susceptibility patterns rather than organismindependent predictions. In summary, the objectives of this research have been successfully achieved: a comprehensive bacteriaspecific dataset was constructed, multiple progressively improved architectures were developed, extensive internal and external evaluations were conducted, and a robust Biophysical Fusion Ensemble was proposed as the final predictive model, providing an effective computational approach for accelerating earlystage antimicrobial peptide screening and supporting future efforts to combat the growing threat of antimicrobial resistance.

6.2 Limitations

The Biophysical Fusion Ensemble that was proposed did well and was better than all the other models. Yet there are still some concerns with the Biophysical Fusion Ensemble that needs to be addressed for further betterment.

6.2.1 Dataset Size and Diversity

One of the concerns with this study is the small size of the training data. After doing a lot of cleaning and checking the quality the final data set with bacteria had 32,607 MIC measurements. This data includes 6,148 peptide sequences from 286 bacterial species.

6.2.2 Dependence on Pretrained Protein Language Models

This research utilizes ESM2T68M to generate contextual protein embeddings for every peptide sequence. Although these embeddings substantially improve predictive performance, the model's effectiveness depends on the quality of the pretrained protein language model.

The ESM model was trained on protein databases using selfsupervised learning instead of predicting antimicrobial activity. Because of this some features that are specific to peptides might not be completely captured in the way the model represents data.

Future protein foundation models that are specially made for studying peptides could offer better ways to represent sequences and help make predictions more accurate.

6.2.3 Computational Requirements

Compared with baseline models, the Biophysical Fusion Ensemble brings much more complicated computing needs. The final architecture combines:

- ESM protein embeddings,
- species embeddings,
- several fusion layers that are entirely coupled,
- a group of five ensembles.

This is because they need more power than the older machine learning methods. Neural networks need a lot of power to work properly. Even though the neural network model works well after it is trained, creating the neural network model and updating the neural network model needs GPU hardware that is up to date.

6.2.4 Absence of Experimental Validation

Computer experiments were used to examine this study in its entirety. These tests included evaluating the system's performance on its own, comparing it to others using the DRAMP benchmark, examining concepts, and determining how well it functioned with various species. These tests demonstrate how well the system predicts outcomes. Nevertheless, no actual laboratory tests were conducted to verify the accuracy of the anticipated MIC values. We would have evidence that the technology is practical and useful for medicinal purposes if we conducted these studies. To find out if the system performs as well as the computer claims, it must be evaluated in a lab.

6.2.5 Limited Biological Property Prediction

Other significant biological characteristics that affect therapeutic appropriateness are not jointly estimated by the current paradigm, which uses MIC values to predict antimicrobial activity. Examples include:

1. cytotoxicity,
2. hemolytic activity,

3. peptide stability,

4. protease resistance,

5. immunogenicity,

6. pharmacokinetic behavior.

A peptide that is really good at fighting off microbes may still not make it as a drug if it has some bad properties. So just because a peptide is good at fighting microbes, that does not mean it will be a drug. The peptide has to have good qualities for it to be a good medication.

6.2.6 Generalization Beyond the Current Dataset

It was the new model. tested utilizing publicly accessible information about specific antibacterial peptides. When we tested our model using both our own and other people's data, it performed well. When we apply the model to newly discovered peptides or fresh data in the future, it could not be as accurate in making predictions. This is due to the possibility that the new peptides or data differ from the ones we used to create the model. The peptides themselves may differ depending on the methods used in the tests or the species that were employed. The model will get even better at producing accurate predictions and function well in a variety of scenarios if we continue to update it with data that we know to be accurate.

Summary

One concept is the Biophysical Fusion Ensemble. It demonstrates how successfully pretrained protein language embeddings, species specific data, and manually created biophysical descriptors work together to predict antibacterial MIC. Out of all the models tested, the Biophysical Fusion Ensemble model performed the best. It establishes a benchmark for next studies on the Biophysical Fusion Ensemble. The Biophysical Fusion Ensemble's shortcomings aren't actually flaws. more opportunities to improve it. This can be accomplished by testing it in actual experiments using the Biophysical Fusion Ensemble and utilizing datasets that include additional species with improved biological representations.

6.3 Future Research Directions

The findings of this study show that an efficient framework for speciesconditioned antibacterial MIC prediction may be created by combining protein language model

embeddings with biophysical descriptors and ensemble learning. Even though the suggested Biophysical Fusion Ensemble outperformed all other models tested, there are still a number of ways to increase forecast accuracy, robustness, and usefulness. Future study should focus on the following areas.

6.3.1 Expansion of HighQuality Training Datasets

An improved set of antibacterial peptides, this work has a bright future. It was found that it is more vital to make sure the information we have is useful and correct than just having large data.

Deep learning models would be able to learn richer sequence–activity connections and enhance their capacity to generalize to previously undiscovered peptide families and bacterial species with a larger and more varied dataset.

6.3.2 Generalization to Unseen Bacterial Species

The existing approach is restricted to species that the model have already seen because it makes use of learnt species embeddings. The model must discover methods to accomplish this, such as creating embeddings using phylogeny or taxonomy or representing species using graphs and metadata. In order to enable the framework to make predictions about species it has never seen before, it may also attempt to employ zeroshot or fewshot learning. As a result, the bacterial species framework will be more practical in daily life. If these can be achieve, the bacterial species framework will be improved. Enhancing the species framework is the primary objective.

6.3.3 Integration of Structural Information

Future study needs to include information, like projected 3D structures, AlphaFold embeddings, residue contact maps, secondary structure, solvent accessibility and membrane interaction properties. The prediction of activity may be enhanced by incorporating sequencebased and structurebased representations.

6.3.4 Exploration of Advanced Protein Foundation Models

The ESM2T68M model was selected in this investigation because it performs well and yields decent results. Future research should examine protein models such as ProtT5, ProtBERT, ESM, and ProGen. These models can improve MIC prediction and provide further insight into how compounds function. The ESM2T68M

model was selected in this investigation because it performs well and yields decent results. Future research should examine protein models such as ProtT5, ProtBERT, ESM, and ProGen. These models can improve MIC prediction and provide further insight into how compounds function.

6.3.5 Improved Ensemble Learning Strategies

Future studies should examine sophisticated ensemble techniques, such as weighted and stacking ensembles, Bayesian ensembles, uncertaintyaware fusion, adaptive weighting, and mixtureofexperts architectures, to further improve accuracy and reliability, even though the fivemember ensemble improved predictive performance.

6.3.6 MultiTask Learning Frameworks

This framework needs to be improved so that it can simultaneously make numerous predictions regarding peptides. This includes how they fend against harmful substances, whether they are poisonous, whether they degrade red blood cells, how stable they are, whether they can resist being broken down by other substances, whether they trigger an immune response, how well they mix with water, and what happens to them in the body. The computer will be more adept at determining if a peptide is beneficial or dangerous if we can educate it to learn all these things. This process, known as task learning, can provide us a clearer understanding of the capabilities of peptides.

6.3.7 Integration with Generative Antimicrobial Peptide Design

A method for discovering novel antimicrobial peptides can be created by combining the new predictor with peptide generating models. This system is capable of creating peptides, predicting their efficacy, determining whether they are hazardous, ensuring that they differ from one another, searching for novel and intriguing ones, and selecting which ones to attempt first. This can speed up our search for peptides. Together, the predictor and the peptide generating models can facilitate this process. This system will speed up the peptide discovery process.

6.3.8 Experimental Validation

Future research should validate the suggested framework by synthesizing specific peptides, calculating their minimum inhibitory concentrations (MICs), comparing

the predicted and experimental activity, and applying the findings to improve the model.

6.3.9 Clinical Translation and Decision Support

Future studies may broaden the framework to support healthcare decisionmaking. This can entail recommending therapies tailored to particular microorganisms. Additionally, it may aid in selecting peptides that are effective despite the bacteria's resistance. It may forecast each person's medication. Additionally, it could determine the combination of medications to be used. It might forecast how well medications will combat bacteria in the future.

6.3.10 Continuous Learning Framework

Future systems should be able to learn continuously as peptide databases grow in size. This implies that they must be able to add validated antimicrobial peptides to the model without forgetting the previous ones. We want the systems to continue functioning throughout time so they can continue to produce accurate forecasts. The systems must be able to learn about antimicrobial peptides and retain what they already know about them since they are crucial.

Summary

The proposed Biophysical Fusion Ensemble provides a robust computational basis for the prediction of antibacterial MICs, but also presents many avenues for further development. The most promising directions for future research include expanding highquality datasets, enhancing the ability of the models to generalize across species, adding structural and biological information, using larger protein foundation models, creating more sophisticated ensemble methods, and combining activity prediction with peptide generation. In conclusion, these developments, coupled with experimental validation and ongoing learning, may pave the way for the creation of comprehensive AIdriven platforms that can expedite the discovery of antimicrobial peptides and contribute to the battle against antimicrobial resistance on a global scale.

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